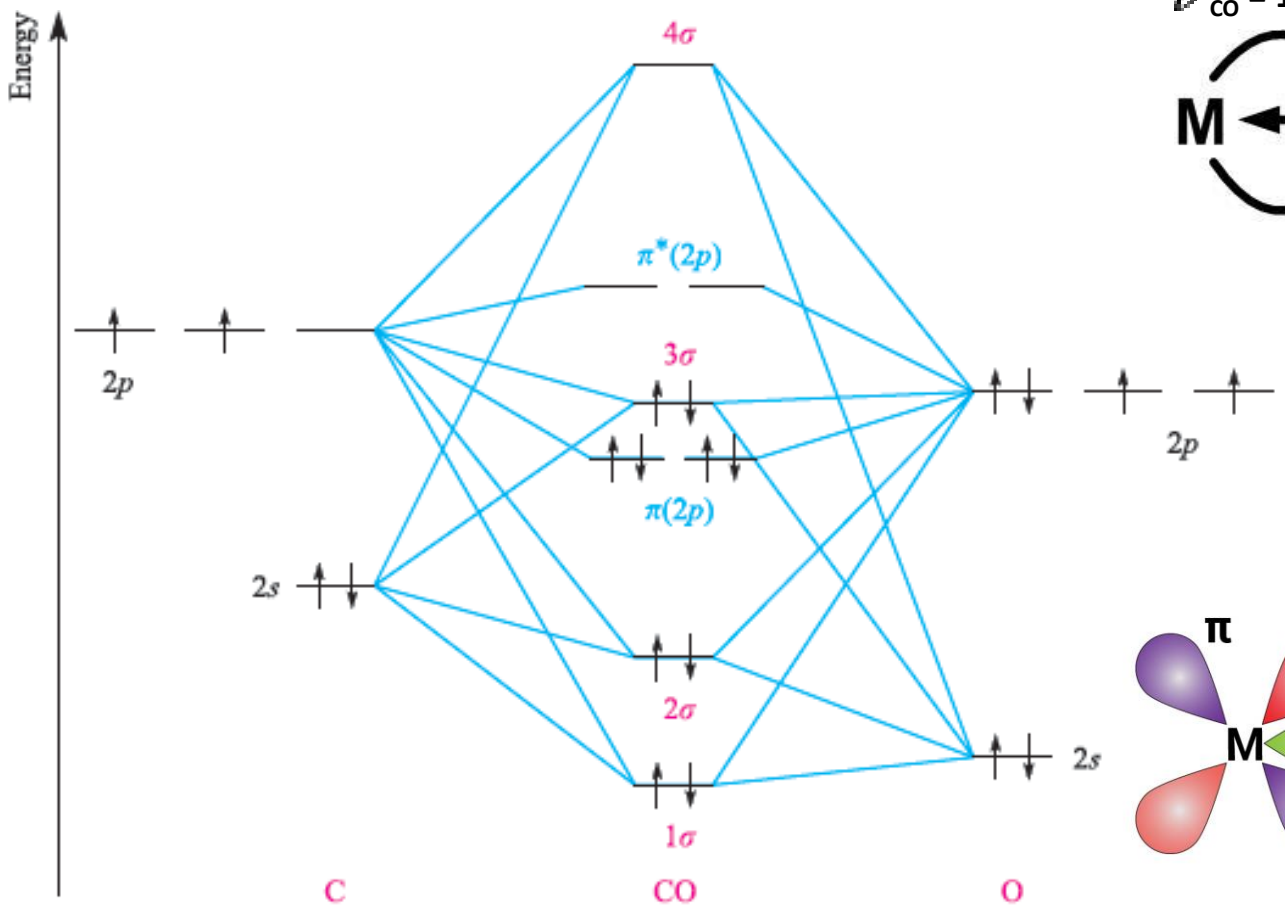
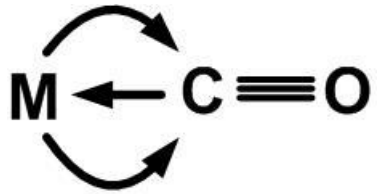
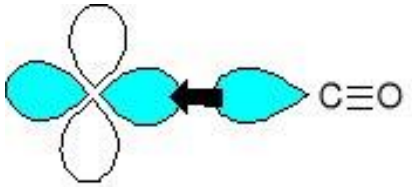
Molecular Orbital Theory



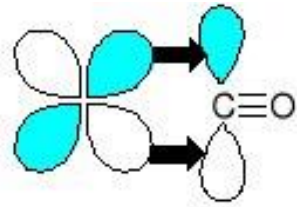
MO diagram

$\tilde{\nu}_{CO}$ = 1670-1820 cm^{-1}

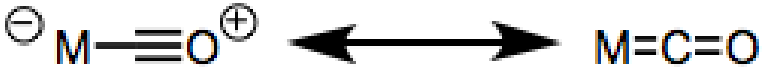




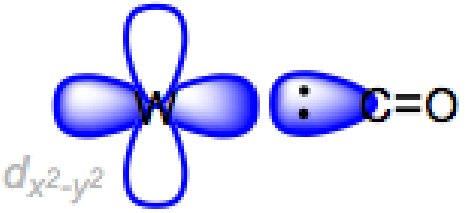
sigma bond



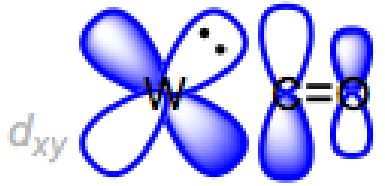
pi back-bond



σ-type orbital interaction

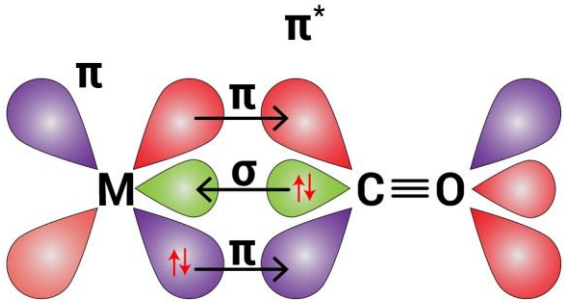


π-type orbital interaction



Electron donor!

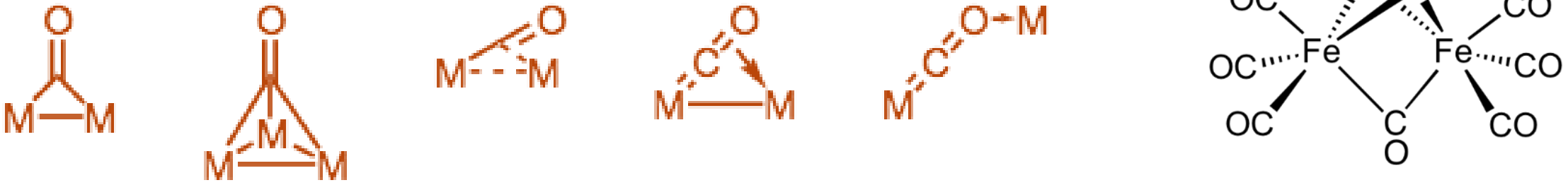
Electron acceptor!



Bond order and bond strength **Free CO = 2143 cm^{-1}**

Carbon Monoxide

Bridging modes of CO

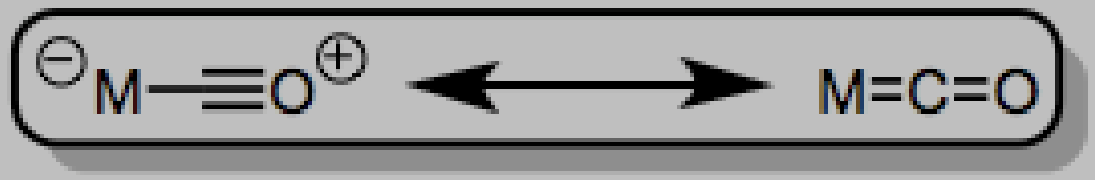


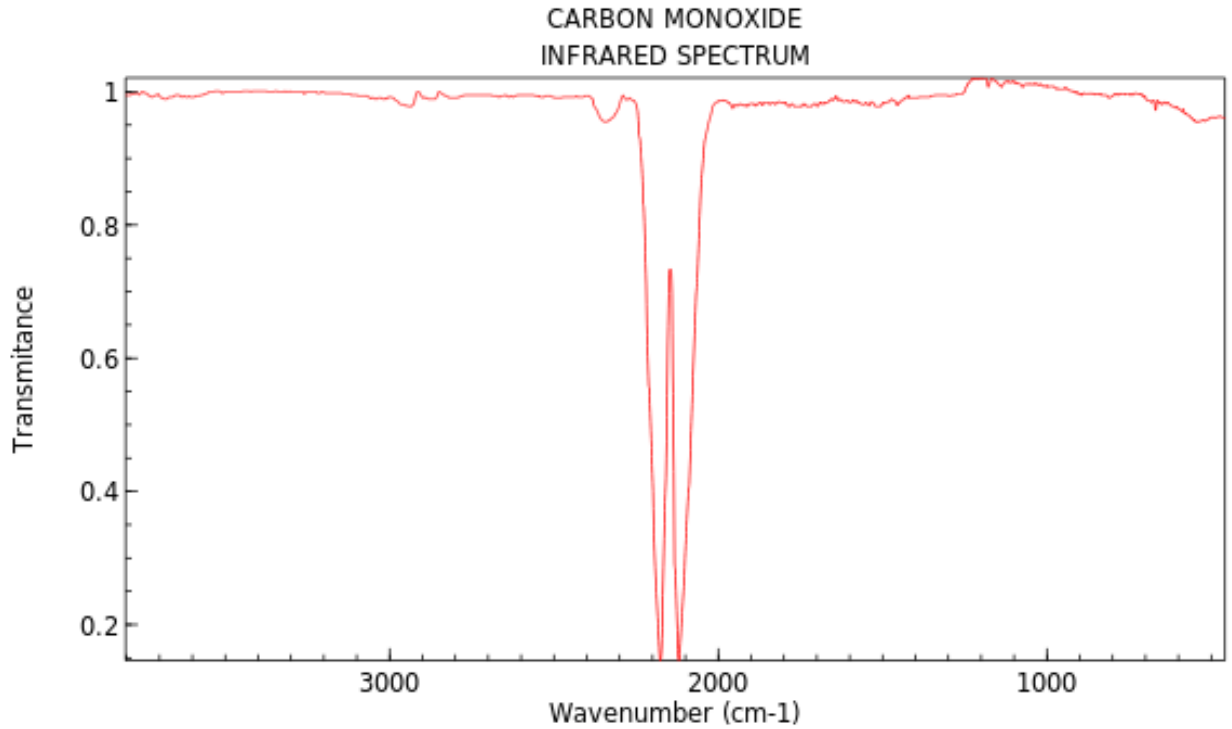
❖ How the wave number of CO will change in metal carbonyl complexes

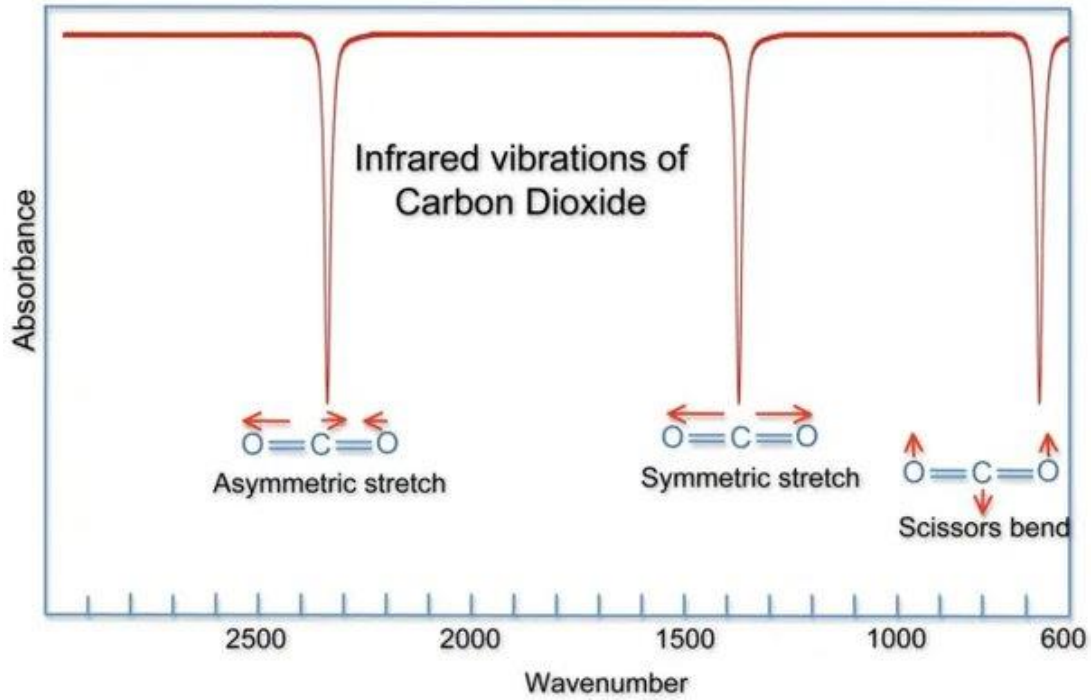
Free CO
2143 cm^{-1}

$V(CO)_6$	$Cr(CO)_6$	$Mn_2(CO)_{10}$	$Fe(CO)_5$	$Co_2(CO)_8$	$Ni(CO)_4$
1976 cm^{-1}	2000 cm^{-1}	2013 cm^{-1}	2023 cm^{-1}	2044 cm^{-1}	2057 cm^{-1}

$[Ti(CO)_6]^{2-}$
1747 cm^{-1}







Mutual Exclusion Principle

For molecules with a center of symmetry, no IR active transitions are Raman active and vice versa

⇒*Symmetric molecules*

IR-active vibrations are not Raman-active.

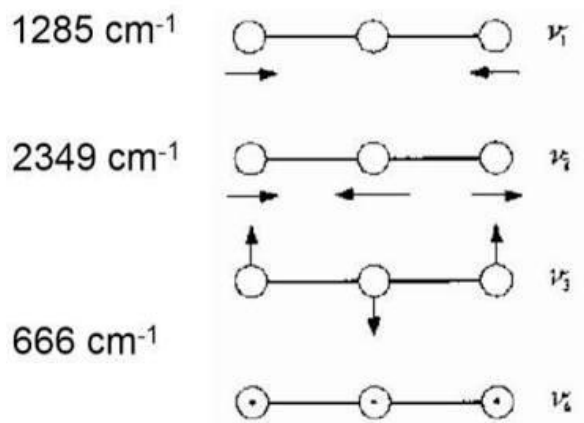
Raman-active vibrations are not IR-active.

O = C = O ← →

Raman active
IR inactive

O = C = O ← ←

Raman inactive
IR active



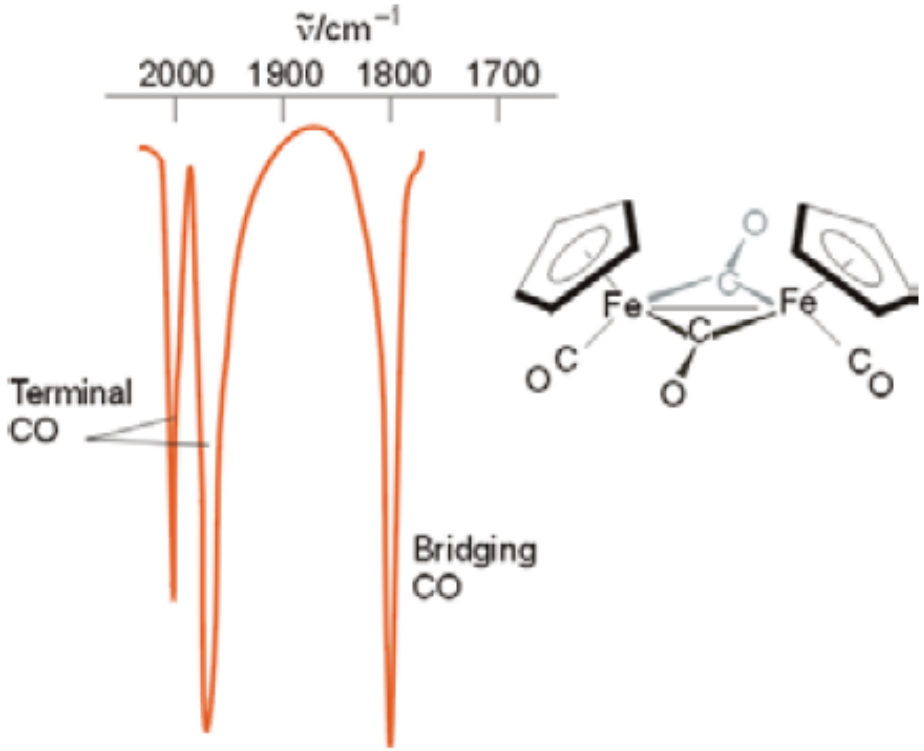
$\nu_{(s)}$ **C=O**
IR inactive, Raman active

$\nu_{(as)}$ **C=O**
IR active, Raman active

$\delta_{(s)}$ **O=C=O**
IR active, degenerated

Example

- IR spectrum shows bands in the bridging and terminal regions
- Bridging IR band is single implying that the two COs are almost co-linear
- Two terminal IR stretch bands implying that the two COs are not co-linear



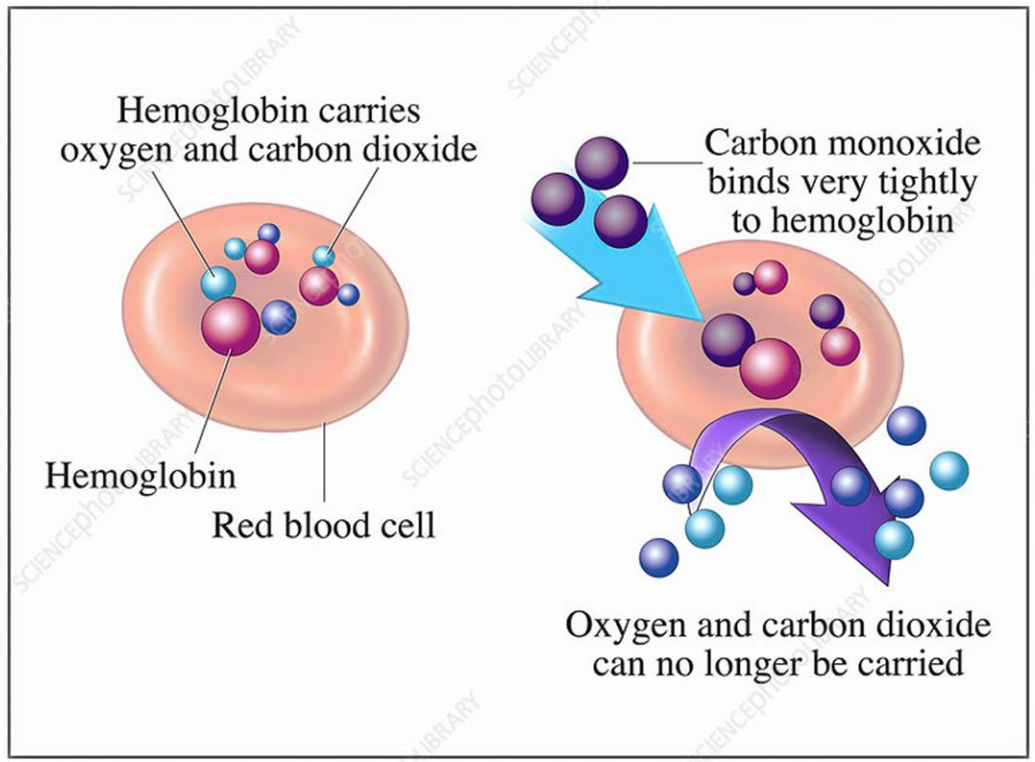
Functional Group	Wavenumber (cm^{-1})	Intensity
Akyl		
C-H	2853-2962	medium-strong
Alkenyl		
=C-H	3010-3095	medium
C=C	1620-1680	variable
Alkynyl		
≡C-H	3300	strong
C≡C	2100-2260	variable
Aromatic		
Ar-H	3030	variable
C=C	1400-1600	variable
Alcohols, Phenols & Carboxylic Acids		
O-H (alcohols / phenols)	3200-3500	broad, strong
O-H (carboxylic acids)	2500-3000	broad, variable
Aldehydes, Ketones, Esters, Carboxylic acids & Amides		
C=O (Aldehydes)	1690-1740	strong
C=O (Ketones)	1680-1750	strong
C=O (Esters)	1735-1750	strong
C=O (Carboxylic acids)	1710-1780	strong
C=O (Amides)	1630-1690	strong
Amines		
N-H	3300-3500	medium

Infrared absorption data

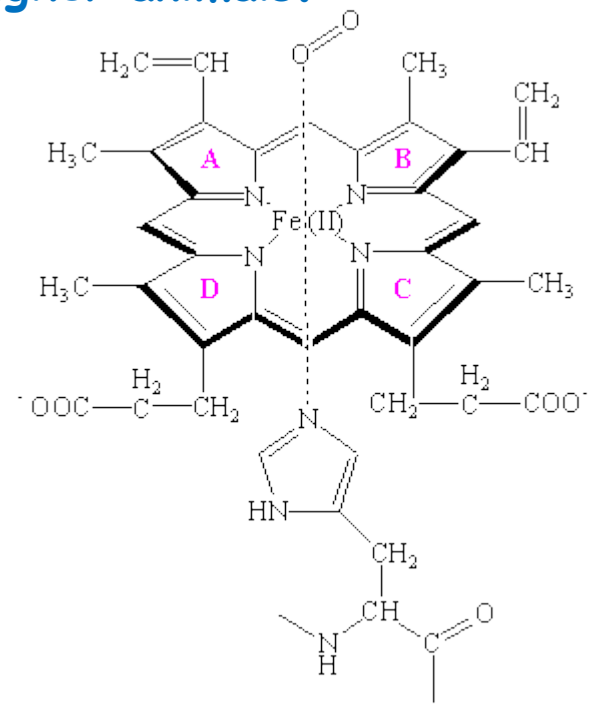
Bond	Wavenumber /cm^{-1}
N—H (amines)	3300 – 3500
O—H (alcohols)	3230 – 3550
C—H	2850 – 3300
O—H (acids)	2500 – 3000
C≡N	2220 – 2260
C=O	1680 – 1750
C=C	1620 – 1680
C—O	1000 – 1300
C—C	750 – 1100

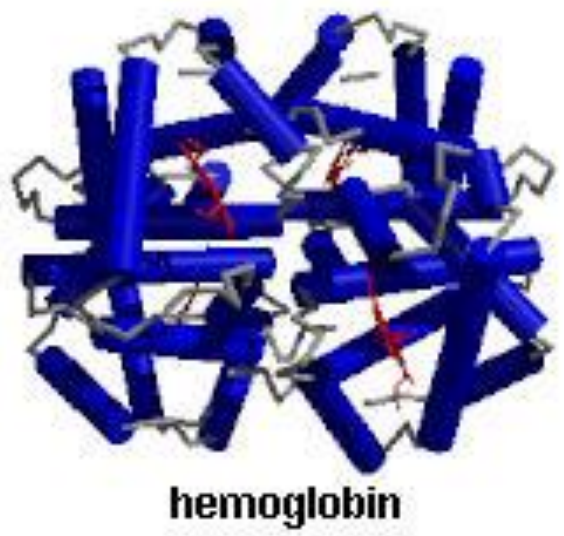
Carbon Monoxide

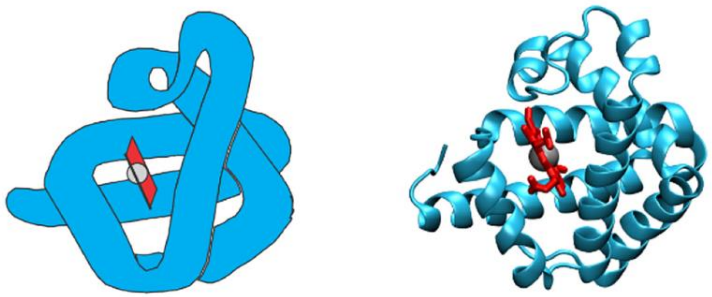
- **Carbon monoxide binds much more strongly to Hb than oxygen – Why?**



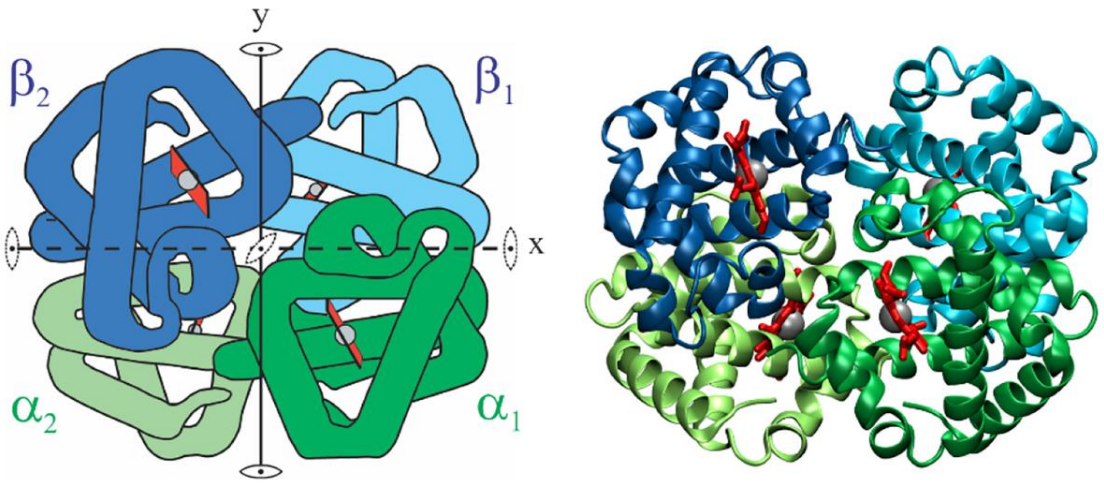
Myoglobin is a hemoprotein, isolated from the cells of vertebrate skeletal muscle that is both a structural and functional relative of **hemoglobin**, the oxygen-transport protein of the blood of higher animals.







myoglobin



hemoglobin

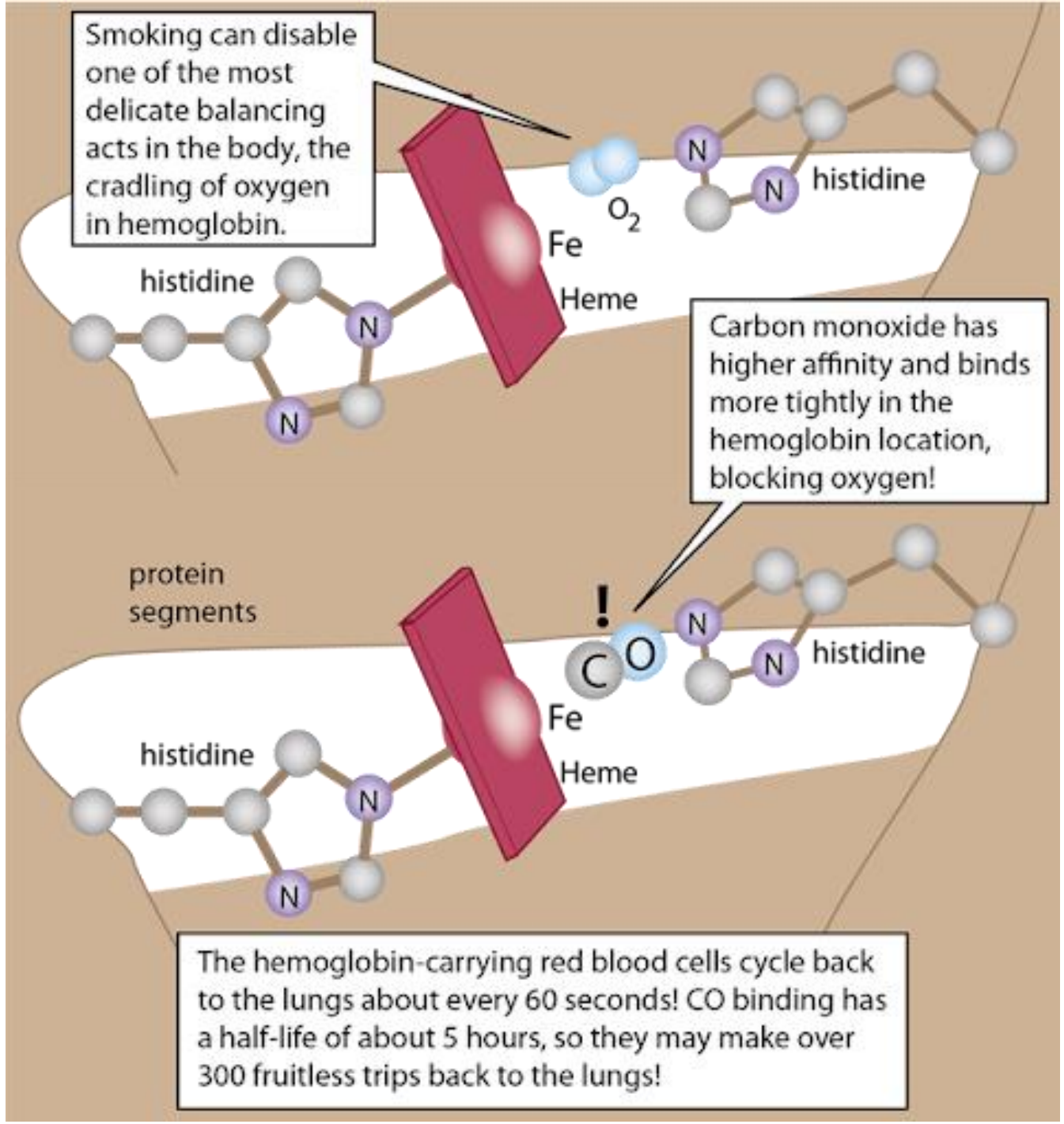
Myoglobin

- It is a monomeric protein that has 153 amino acids
- It has 8 alpha helices
- **Myglobin** = heme group (contains iron) + Globin protein
- Have a linear chain of amino acids
- It has a globular structure

Hemoglobin

- Quaternary structure
- Four subunits: 2 alpha and 2 beta (Adult hemoglobin)
- ***Alpha subunit:*** 141 amino acid, chromosome 16
- ***Beta, gamma and delta subunit:*** 146 amino acid, chromosome 11

- Each subunit or polypeptide has a heme prosthetic group bound
- Each Hb has four heme
- Each heme Fe^{2+} binds to single oxygen molecule
- Therefore each Hb can carry four oxygen molecules


Smoking can disable one of the most delicate balancing acts in the body, the cradling of oxygen in hemoglobin.
O_2
N
N
histidine
Fe
Heme
histidine
N
N
Carbon monoxide has higher affinity and binds more tightly in the hemoglobin location, blocking oxygen!
protein segments
!
C
O
N
N
histidine
Fe
Heme
histidine
N
N
The hemoglobin-carrying red blood cells cycle back to the lungs about every 60 seconds! CO binding has a half-life of about 5 hours, so they may make over 300 fruitless trips back to the lungs!

Carbon Monoxide

- At the center of the heme is an Fe(II) atom. Four of the six coordination sites around this atom are occupied by nitrogen atoms from a planar porphyrin ring

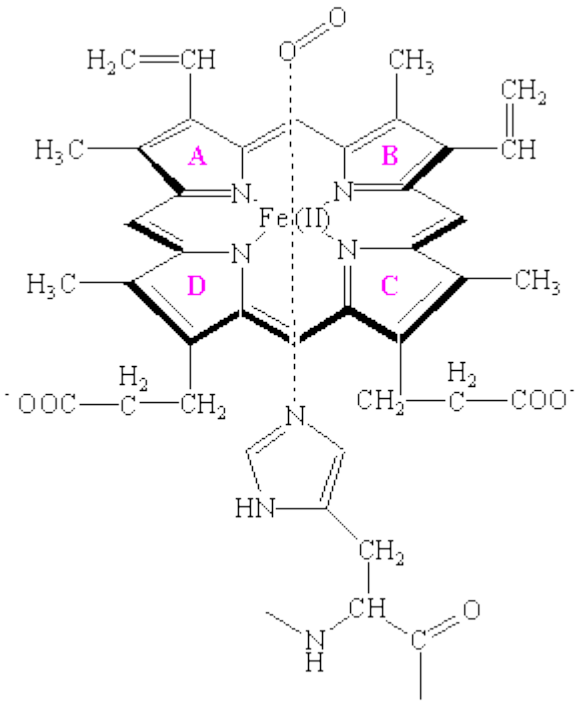
- The fifth coordination site is occupied by a nitrogen atom from a histidine side chain on one of the amino acids in the protein

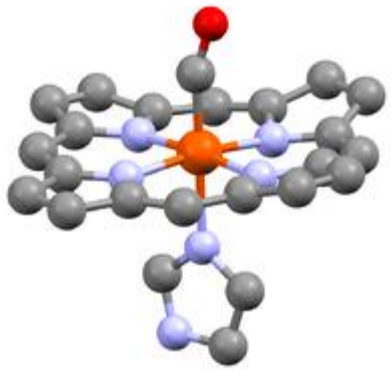
- The last coordination site is available to bind an O_2 molecule

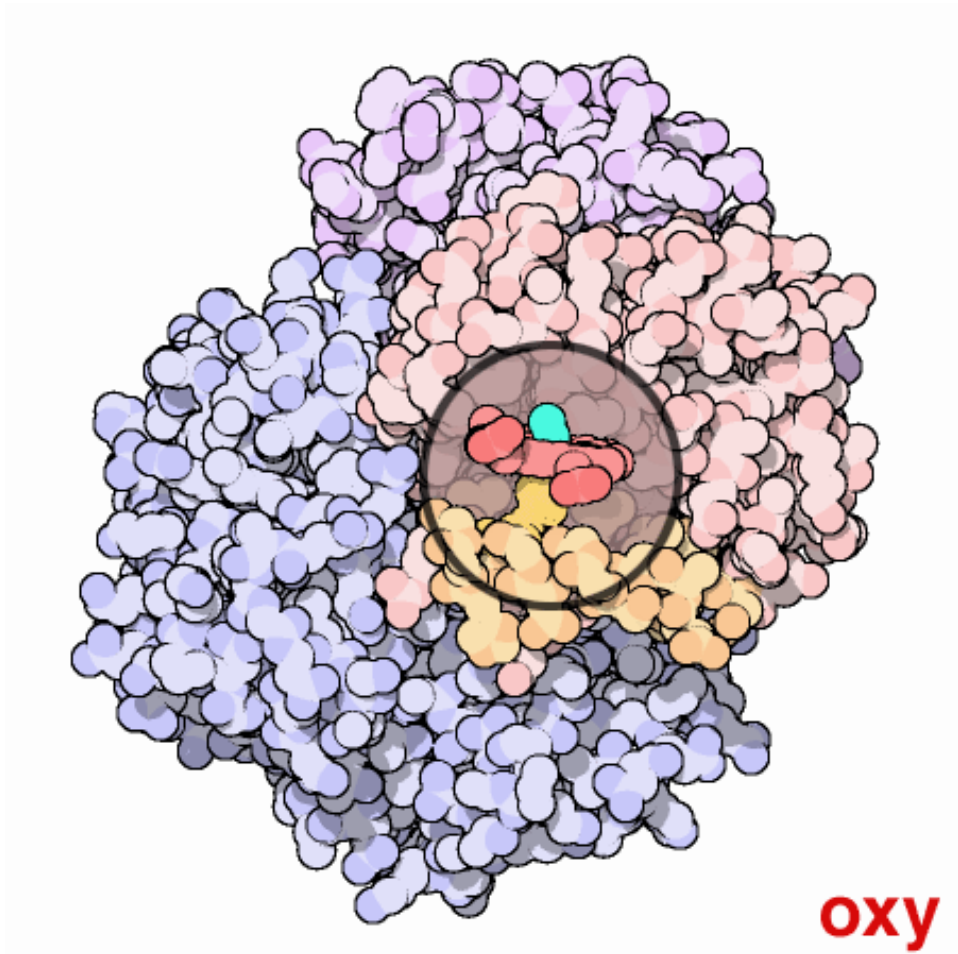
- The heme is therefore the oxygen-carrying portion of the hemoglobin and myoglobin molecules

- The structure of myoglobin suggests that the oxygen-carrying heme group is buried inside the protein portion of this molecule, which keeps pairs of hemes group from coming too close together

- This is important, because these proteins need to bind O_2 reversibly and the Fe(II) heme, by itself, cannot do this. When there is no globin to protect the heme, it reacts with oxygen to form dimers.





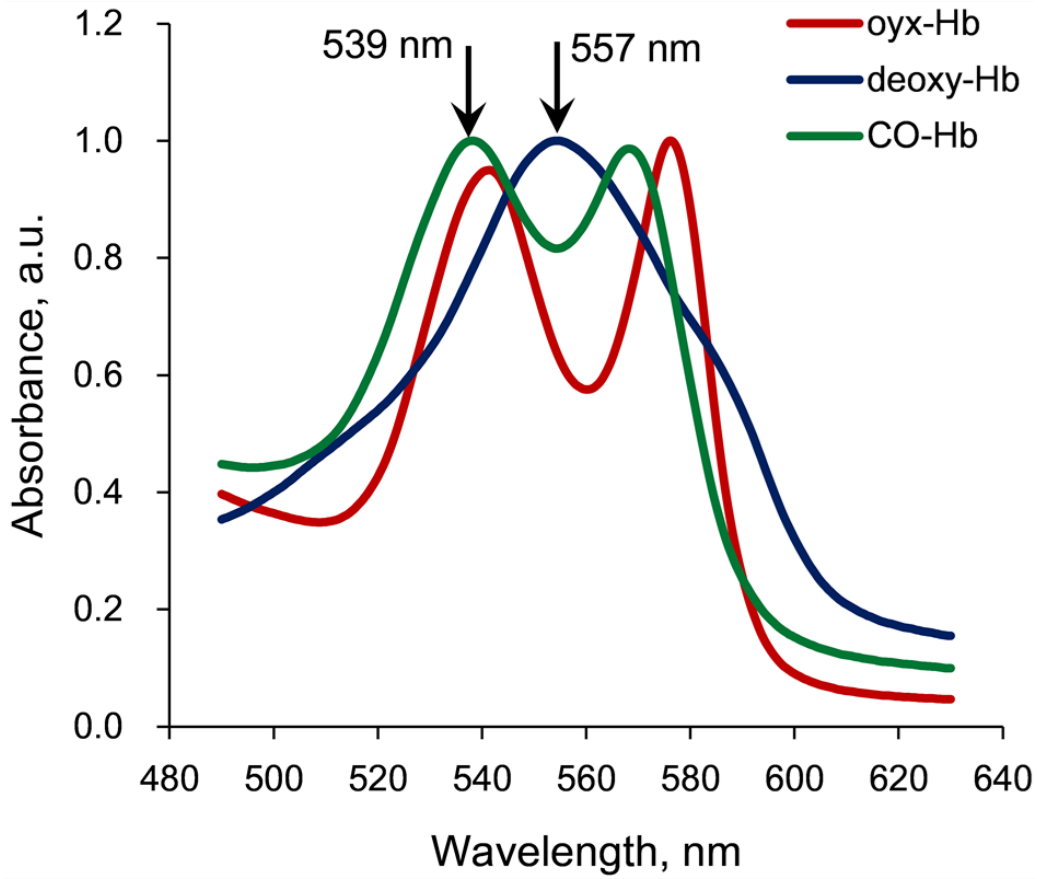

oxy

Preferential Binding of CO and deactivation of Hb and Mb

$Mb(O_2) + CO \rightleftharpoons Hb(CO) + O_2$

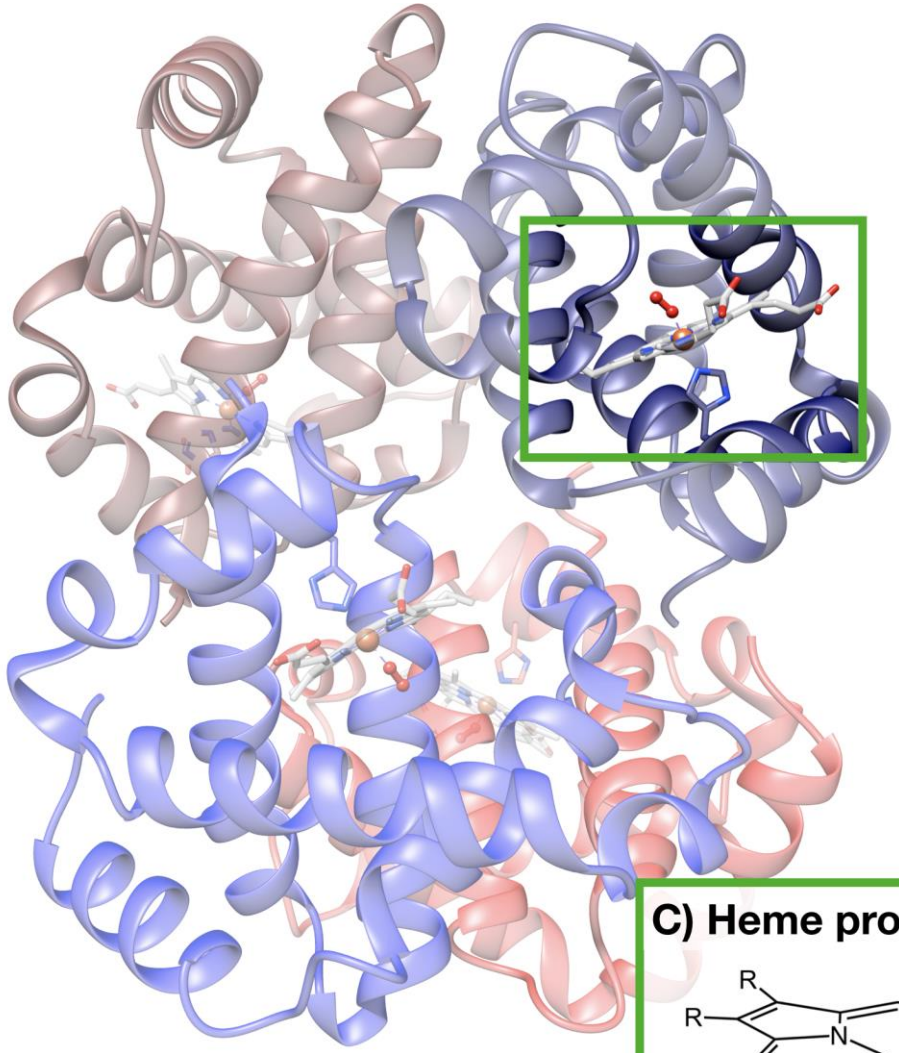
Carboxyhemoglobin

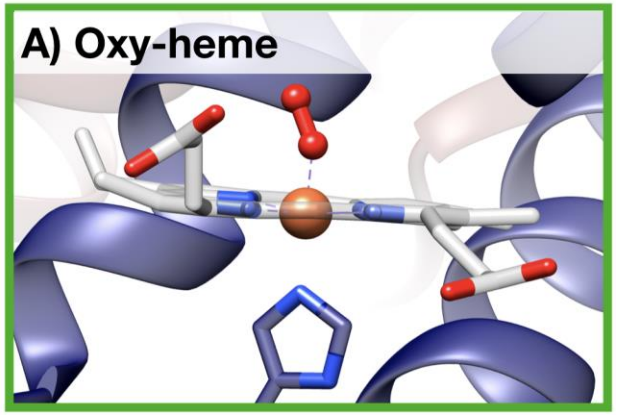
Binding affinity of free Heme group for CO is about 25,000 times than O_2

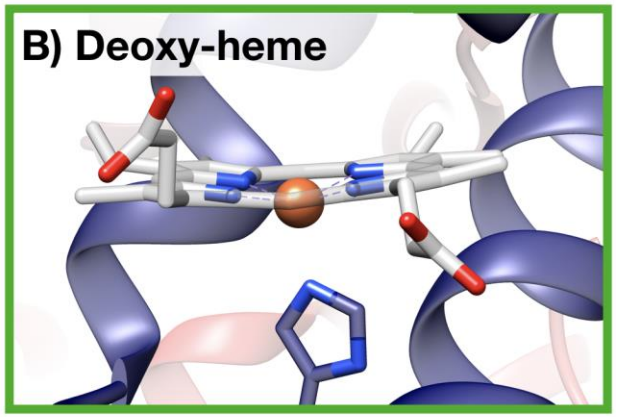


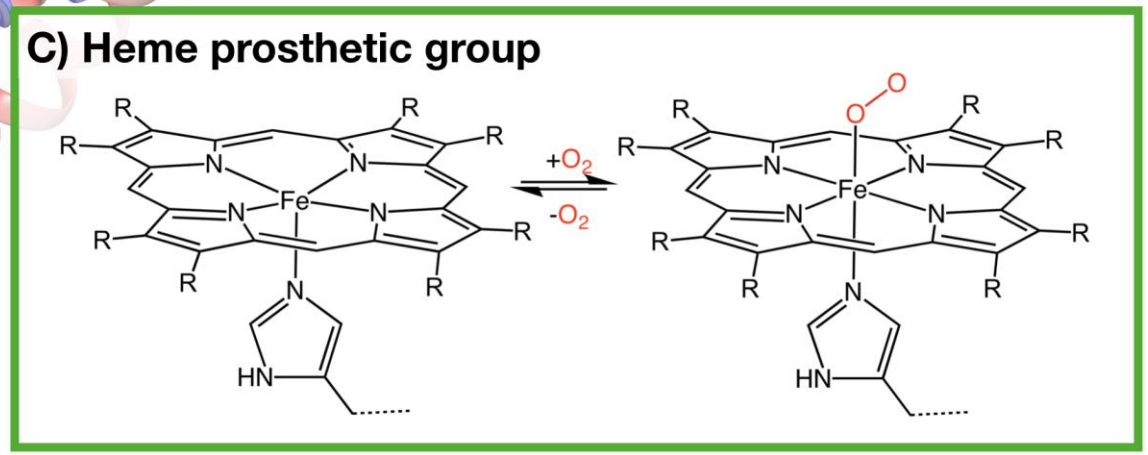
Oxygen Binding to Mb and Hb

- **Oxymyoglobin** - oxygen bearing myoglobin
- **Deoxymyoglobin** - oxygen-free myoglobin
- In oxymyoglobin, six ligands are coordinated to the ferrous ion in octahedral symmetry
- Oxygen is coordinated between the iron and the imidazole sidechain of His-64




A) Oxy-heme


B) Deoxy-heme


C) Heme prosthetic group
R
N
Fe
HN
+O2
-O2
O
O

MYOGLOBIN

- Myoglobin (Mr 16,700; abbreviated Mb) is a relatively simple oxygen-binding protein found in almost all mammals, primarily in muscle tissue
- Mb is a single polypeptide chain with one molecule of heme. Human Mb contains 152 amino acids with a molecular weight of 17,500 Daltons. One molecule of Mb can combine with 1 molecule of oxygen.
- In the muscles, the oxygen is taken up by Mb for the sake of tissue respiration.

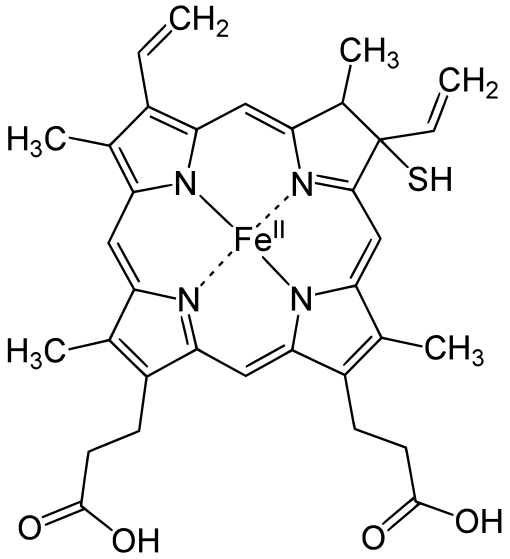
Myoglobin and Hemoglobin

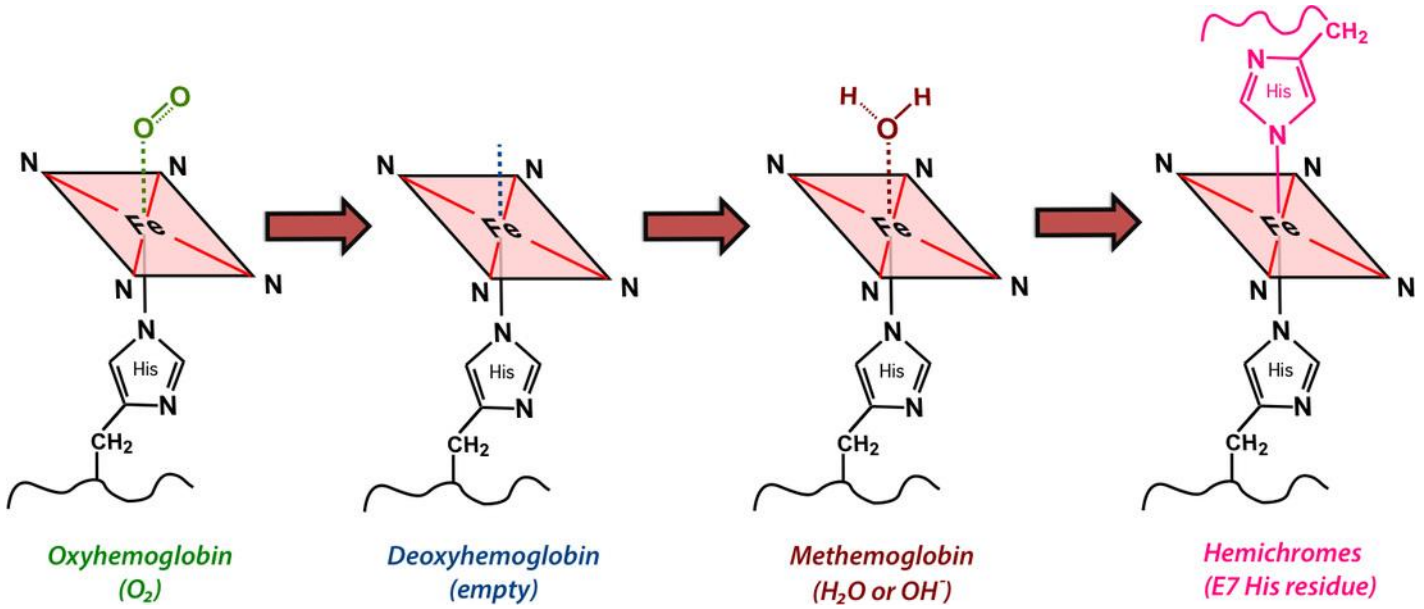
- **Myoglobin (Mb)**
  - Increases O_2 solubility in tissues (muscle)
  - Facilitates O_2 diffusion
  - Stores O_2 in tissues
- **Hemoglobin (Hb)**
  - Transports O_2 from lungs to peripheral tissues (erythrocytes)

Derivatives of hemoglobin

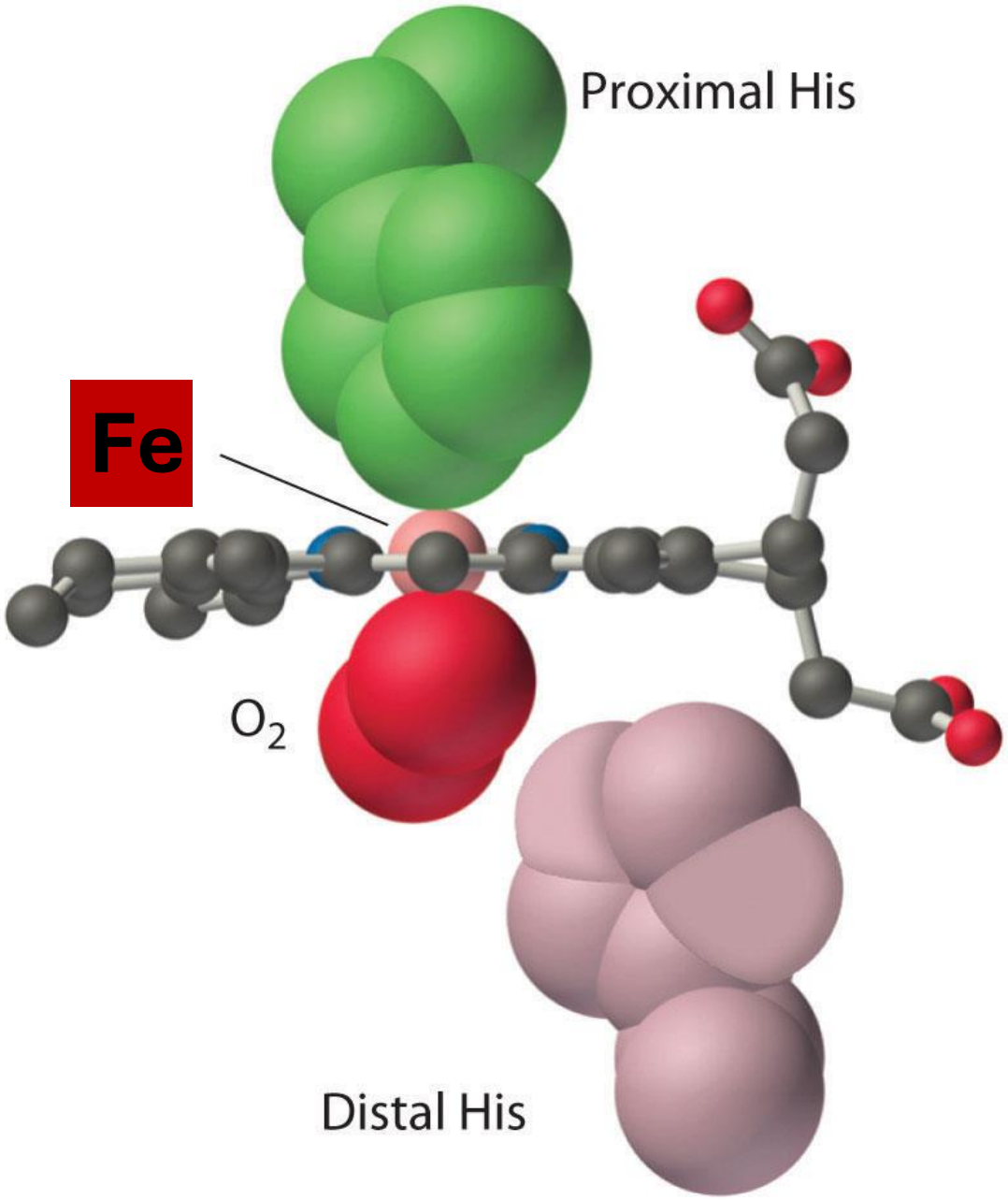
- **Oxyhemoglobin** (oxyHb) = Hb with O_2
- **Deoxyhemoglobin** (deoxyHb) = Hb without O_2
- **Methemoglobin** (metHb) contains Fe^{3+} instead of Fe^{2+} in heme groups
- **Carbonylhemoglobin** (HbCO) – **CO binds to Fe^{2+}** in heme in case of CO poisoning or smoking. CO has 200x higher affinity to Fe^{2+} than O_2.
- **Carbaminohemoglobin** ($HbCO_2$) - CO_2 is non-covalently bound to globin chain of Hb. **$HbCO_2$ transports CO_2 in blood** (about 23%).
- **Glycohemoglobin** (HbA1c) is formed spontaneously by nonenzymatic reaction with Glc. People with DM have more HbA1c than normal (> 7%). Measurement of blood HbA1c is useful to get info about long-term control of glycemia.

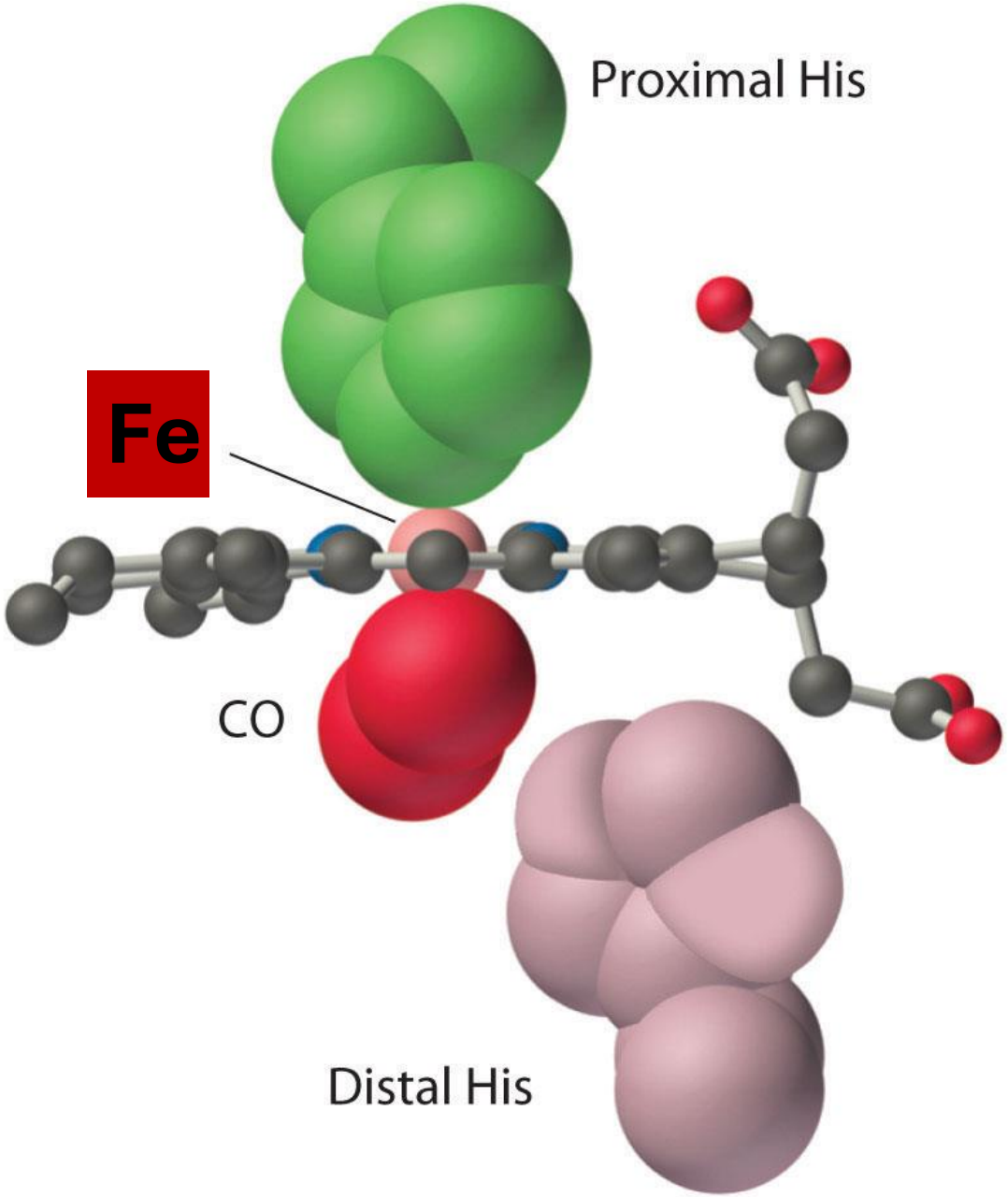
Normal Derivatives	Abnormal Derivatives
1. Oxyhemoglobin	1. Methemoglobin
2. Deoxyhemoglobin	2. Carboxyhemoglobin
3. Carbaminohemoglobin	3. Sulfhemoglobin
	4. Acid hematin



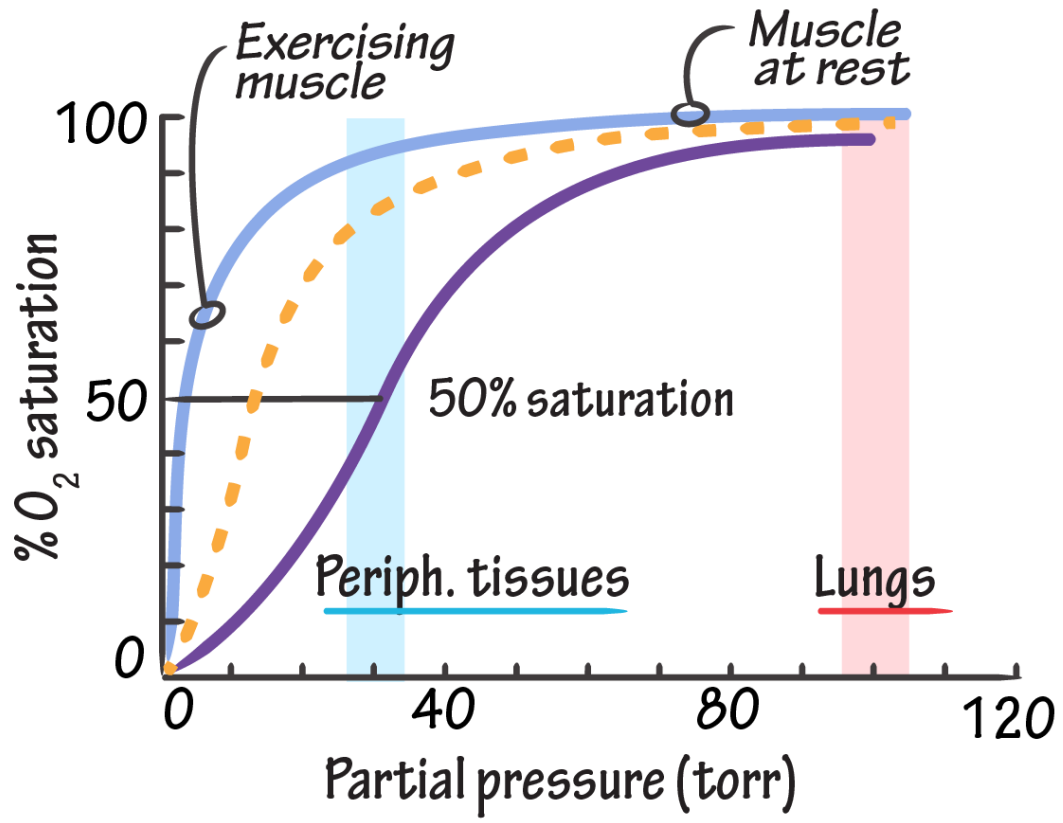


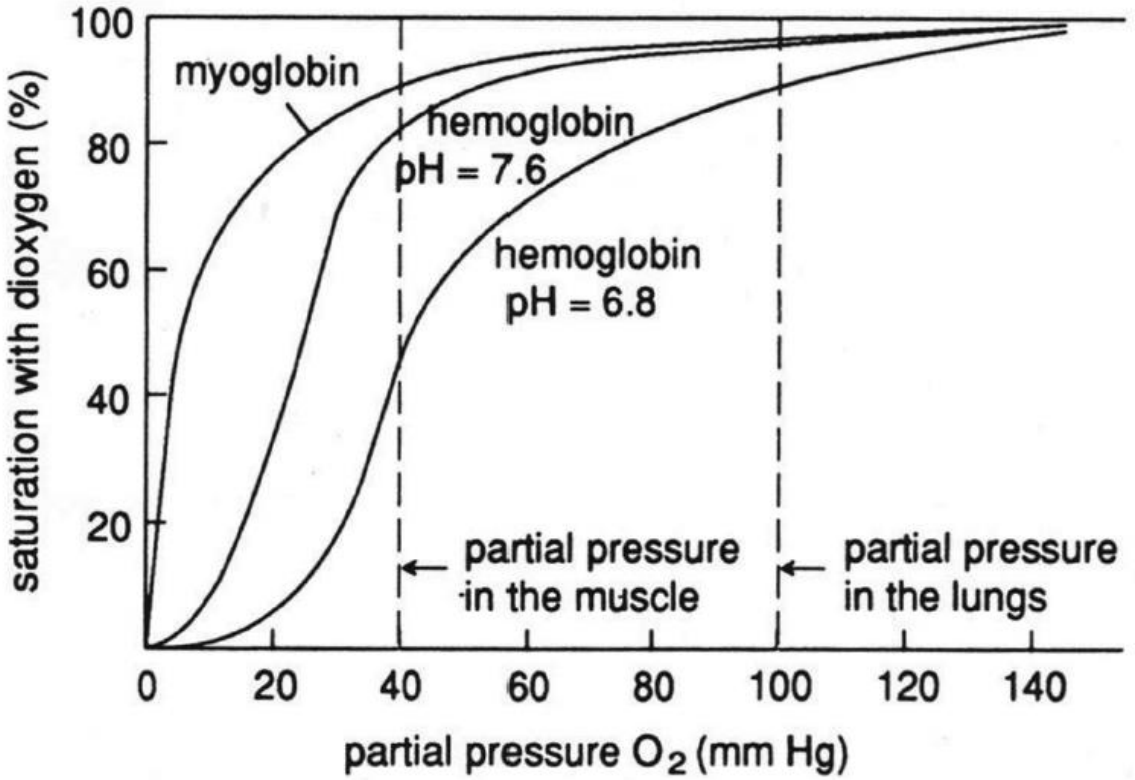



Proximal His
Fe
O_2
Distal His


Proximal His
Fe
CO
Distal His

Dissociation Curves





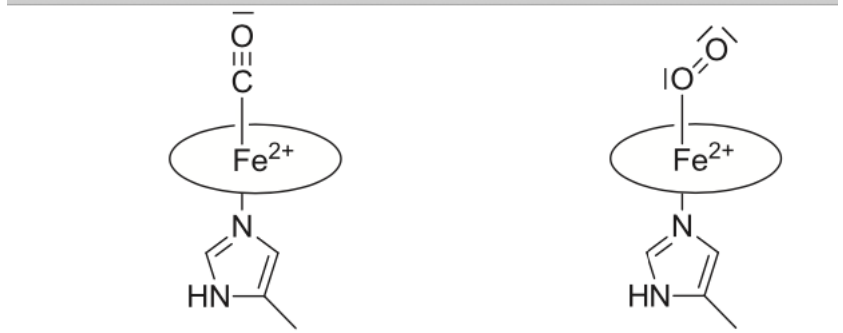
Oxygen saturation curves of myoglobin and hemoglobin at different pH values

Myoglobin

- Higher O_2 affinity
- Responds to muscle's O_2 needs

Hemoglobin

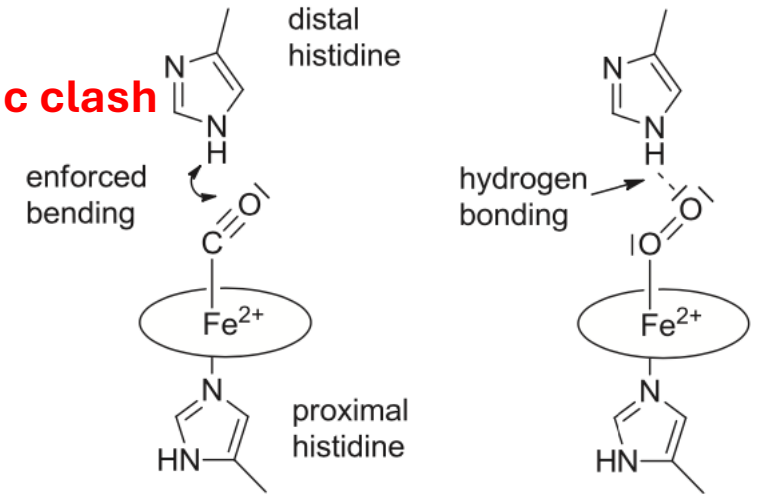
- Cooperative binding = sigmoidal curve
- Responds to O_2 availability



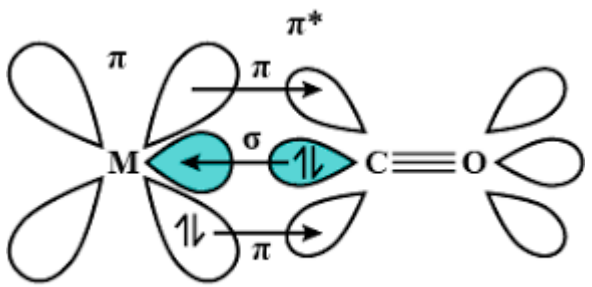
myoglobin, hemoglobin

Protein part allows easier access to small molecules like dioxygen to bind with Fe atoms

Steric clash



Linear coordination of CO is not possible



Synergic bonding in metal carbonyls

Fe^{2+} in low spin

Bonding is weak when CO is bound in angular fashion

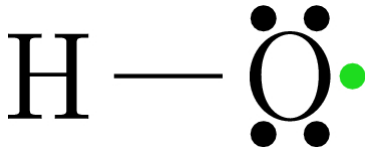
Carbon Monoxide

Fate of Atmospheric CO

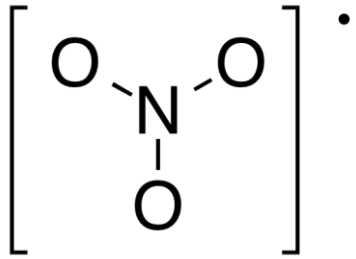
Most of the CO molecules are removed from the atmosphere by oxidation (natural) in the stratosphere.

The most important oxidizing species in the air

- Hydroxyl radical OH

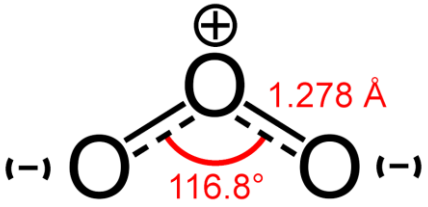


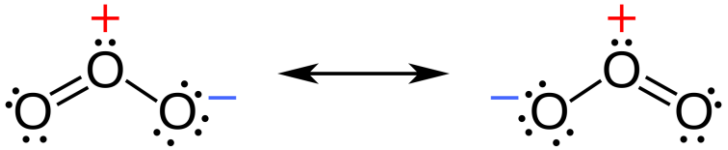
- Nitrate radical NO_3



'detergent of the atmosphere'.

- Ozone molecule O_3





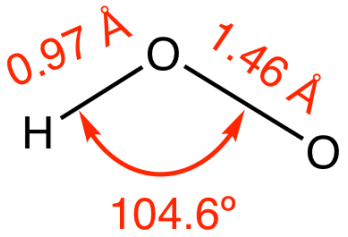
Carbon Monoxide

Fate of Atmospheric CO

It is generally believed that CO is removed from the atmosphere by a reaction with hydroxyl radical.

Oxidized by reaction with hydroxyl radical

- **$CO + HO^{\bullet} \rightarrow CO_2 + H$** **(stratosphere)**



Systematic IUPAC name
Dioxidanyl

- **CO is oxidized to CO_2 in the stratosphere, where it migrates *via* convection, turbulence, and mixing**

- **The reaction is fast and independent of temperature; thanks to it, the residence time of CO in the atmosphere is relatively short, from 2 weeks to 3 months.**

Carbon Monoxide

Production of hydroperoxyl radical

- $O_2 + H \rightarrow HOO\bullet$

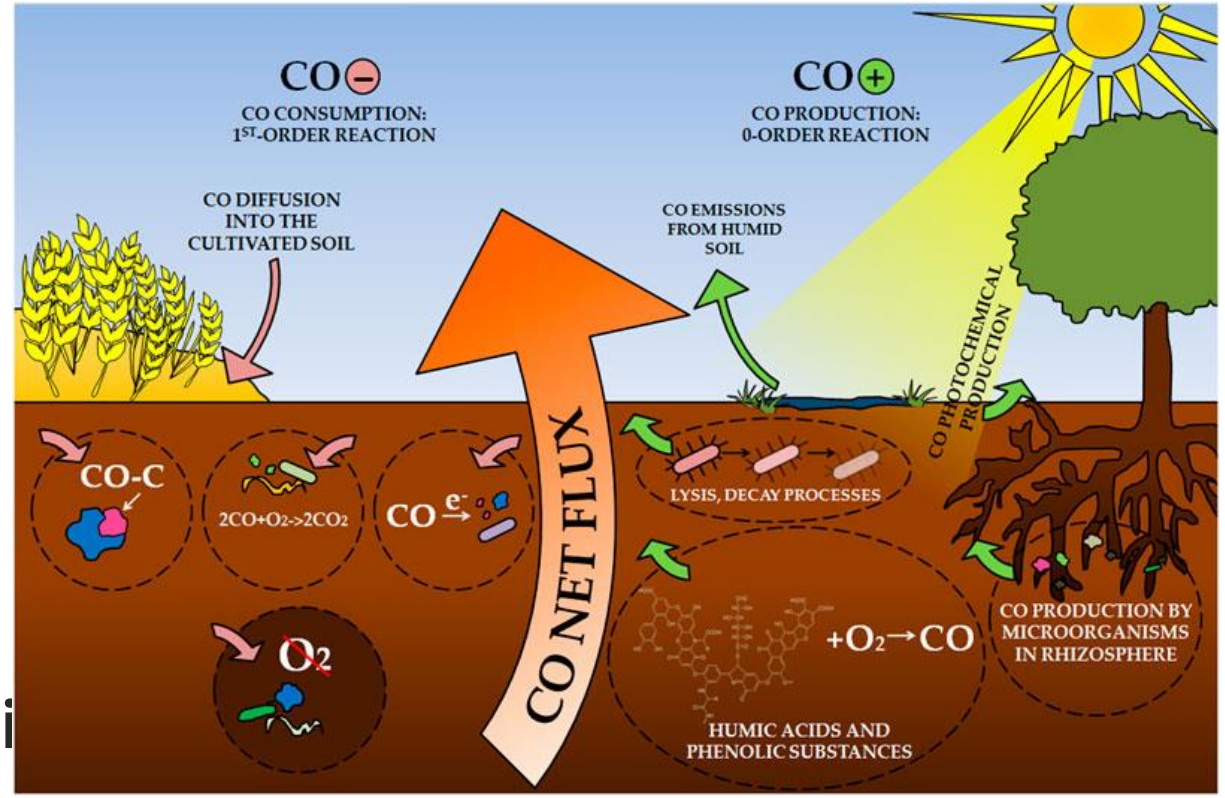
Additional reactions

- $HOO\bullet + NO \rightarrow HO\bullet + NO_2$
- $HOO\bullet + HOO\bullet \rightarrow H_2O_2 + O_2$
- $H_2O_2 + h\nu \rightarrow HO\bullet$

$O_3 + h\nu$ (sun) → $O + O_2$ (wavelength <310 nm)

$O + O_2 + M \rightarrow O_3 + M$

$O + H_2O \rightarrow 2OH$ 3%



CO Fate in Soi

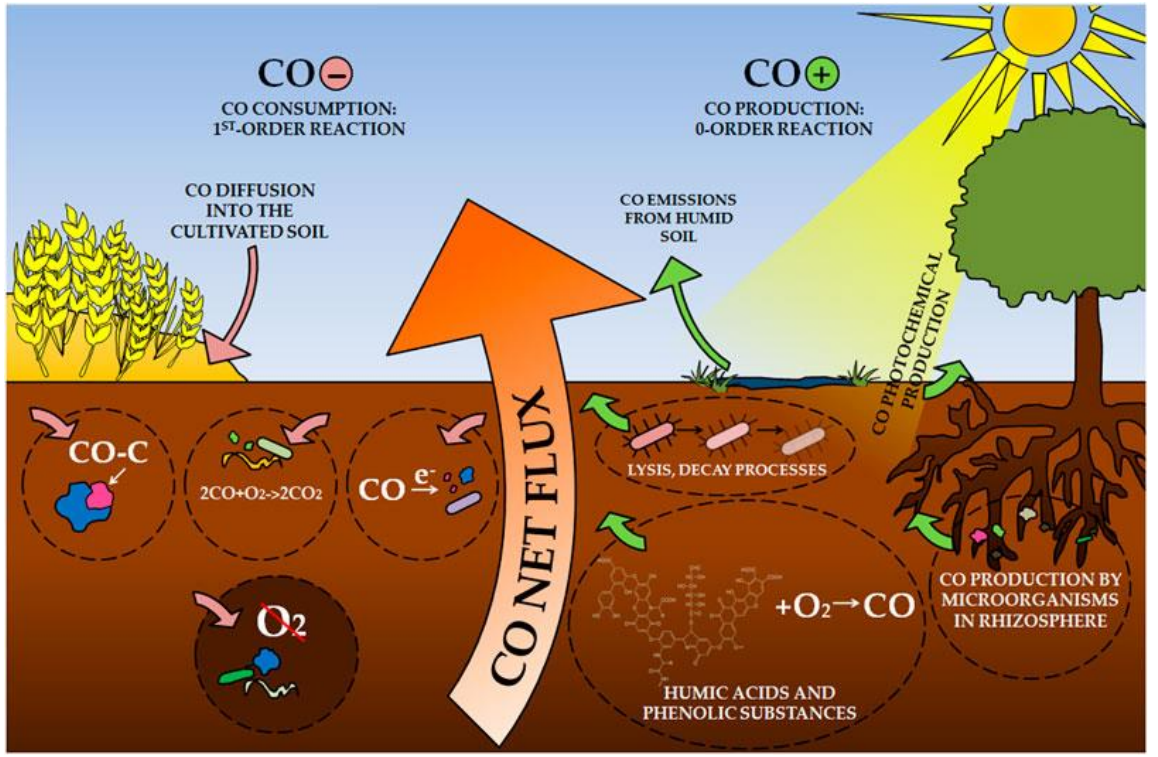
- **Soil is considered one of the main sinks of atmospheric CO, responsible for 40% of total consumption**
- **Soil is a sink for CO**

- **Soil microorganisms metabolize CO**

Humic acid has the average chemical formula $C_{187}H_{186}O_{89}N_9S_1$

The most important soil parameters

- Water Content
- Temperature
- Organic matter content
- pH
- Soil type
- The depth of CO consumption horizon, and CO concentration in the gas phase

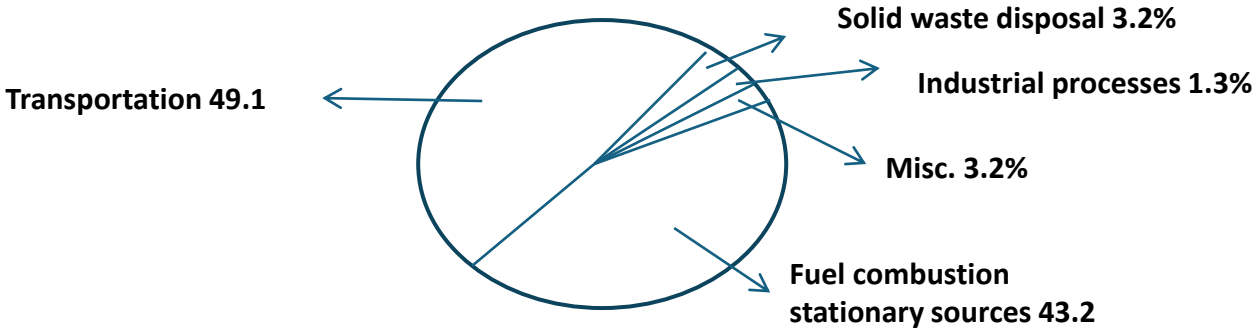


Even small changes in this balancing between CO production and soil uptake can severely impact tropospheric chemistry

Nitrogen Oxides (NO_x)

- **Nitrogen dioxide (NO_2) is the major pollutant in the atmosphere and is formed from nitric oxide (NO). Collectively nitrogen oxides are designated as NO_x**
- **Source of NO_x**
- **Anthropogenic NO_x molecules enter the atmosphere from the combustion of fossil fuels by automobiles, aircraft, and power plants.**
- **At normal atmospheric temp. N_2 and O_2 do not react. However, at very high temp. atmospheric N_2 reacts with oxygen to give a series of complex reactions. Two gases combine to give NO**

$$N_2 + O_2 \longrightarrow 2\,NO \qquad 2NO + O_2 \longrightarrow 2NO_2$$



Nitrogen Oxides (NO_x)

- When NO is released, it combines rapidly with atmospheric oxygen to form NO_2. NO has an unpaired electron and can be written as $NO^{\bullet}$

  - The bacterial decomposition of nitrogen-containing organic matter in soil is another natural source of NO_x

❖ Fate of Atmospheric NO_x

1. Regardless of its source, NO_2 is ultimately removed from the atmosphere as nitric acid and nitrates in dust and rainfall. In a series of complex reactions involving hydroxyl radicals, NO_2 combines with water vapor to form nitric acid. The simplified overall reaction is

$$4\,NO_2 + 2\,H_2O + O_2 \longrightarrow 4\,HNO_3$$

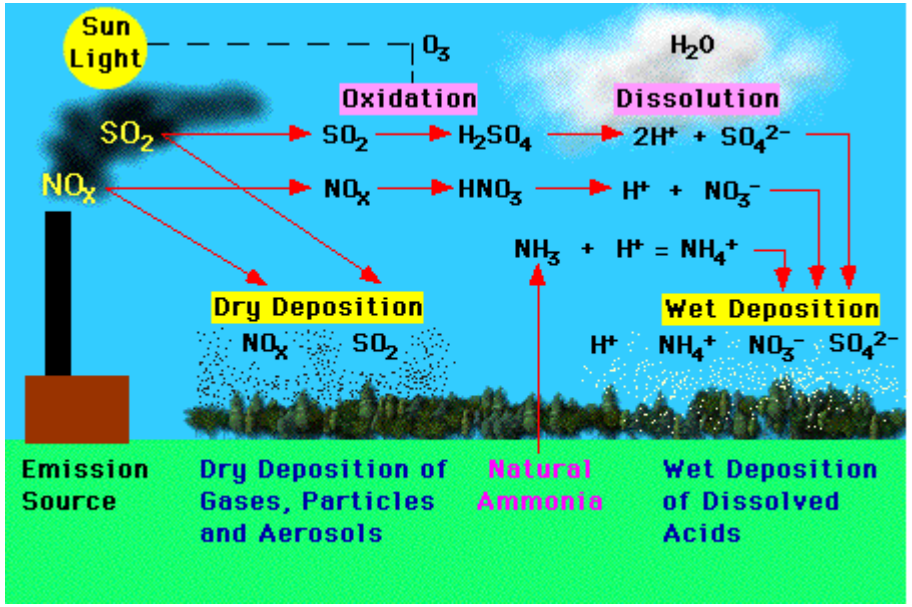
2. Much of the nitric acid in the atmosphere is formed within aqueous aerosols. If the weather conditions are right, the aerosols coalesce into larger droplets in clouds and the result is **acid rain.**

❖ Fate of Atmospheric NO_x

- ❖ **Some of the nitric acid formed reacts with ammonia and metallic particles in the atmosphere to form nitrates**

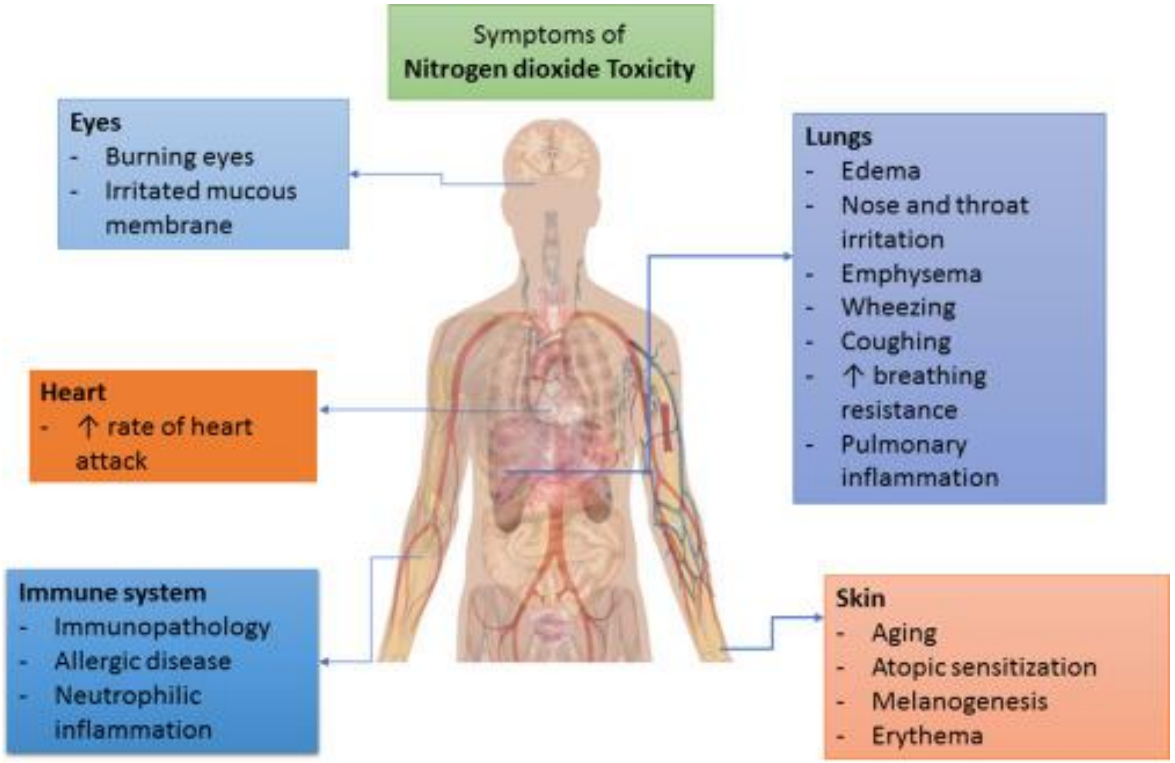
$$HNO_3 + NH_3 \longrightarrow NH_4NO_3$$

- ✓ **Nitrates dissolve in rain and snow or settle as particles. The combined fallout contributes to acid deposition**



Effect of NO_x on Human Health and Environment

- NO_2 is red brown toxic gas and has unpleasant acrid odor. It can cause irritation of the eyes, inflammation of lung tissue, and emphysema.
- NO_x is a serious health problem because of its role in the formation of secondary pollutants associated with photochemical smog.
- Most of the research has concentrated on reducing automobile emission by means of the catalytic converter



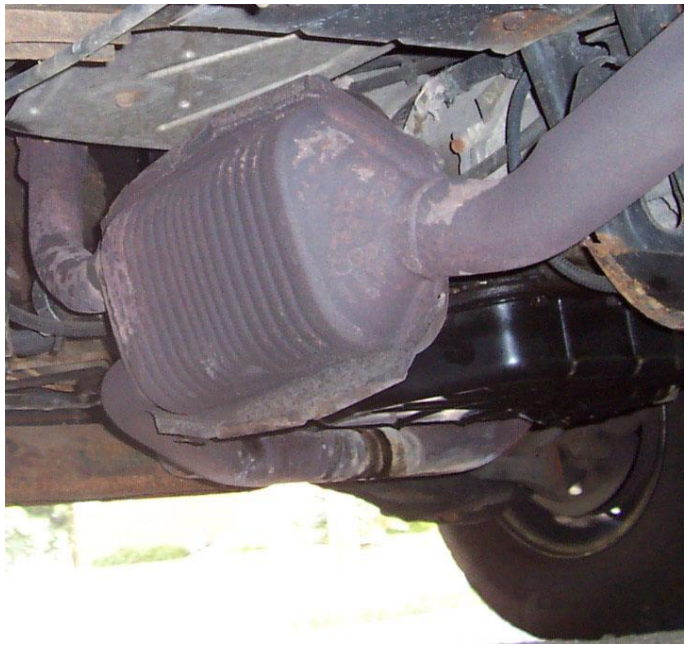
Emphysema is a lung disease that damages the **alveoli** in your lungs. The primary symptom of emphysema is shortness of breath.

Erythema is redness of the skin caused by injury or another inflammation-causing condition

Melanogenesis is the production of the melanin pigments

Atopic sensitization is defined as a positive allergen-specific serum immunoglobulin E (IgE) test or skin prick test to any food or inhalant allergens. It is an important risk factor for the development of asthma and allergic diseases

Catalytic Converter



Reduction of nitrogen oxides to *nitrogen* (N_2)

- $\mathrm{C} + 2\mathrm{NO_2} \rightarrow \mathrm{CO_2} + 2\mathrm{NO}$
- $\mathrm{CO} + \mathrm{NO} \rightarrow \mathrm{CO_2} + \frac{1}{2}\mathrm{N_2}$
- $2\mathrm{CO} + \mathrm{NO_2} \rightarrow 2\mathrm{CO_2} + \frac{1}{2}\mathrm{N_2}$
- $\mathrm{H_2} + \mathrm{NO} \rightarrow \mathrm{H_2O} + \frac{1}{2}\mathrm{N_2}$

Oxidation of carbon, hydrocarbons, and carbon monoxide to carbon dioxide

- $\mathrm{C} + \mathrm{O_2} \rightarrow \mathrm{CO_2}$
- $\mathrm{CO} + \frac{1}{2}\mathrm{O_2} \rightarrow \mathrm{CO_2}$
- $a\,\mathrm{C}_x\mathrm{H}_y + b\,\mathrm{O_2} \rightarrow c\,\mathrm{CO_2} + d\,\mathrm{H_2O} \qquad a, b, c, d, x, y \in \mathbb{Z}$

Thermoelectric materials

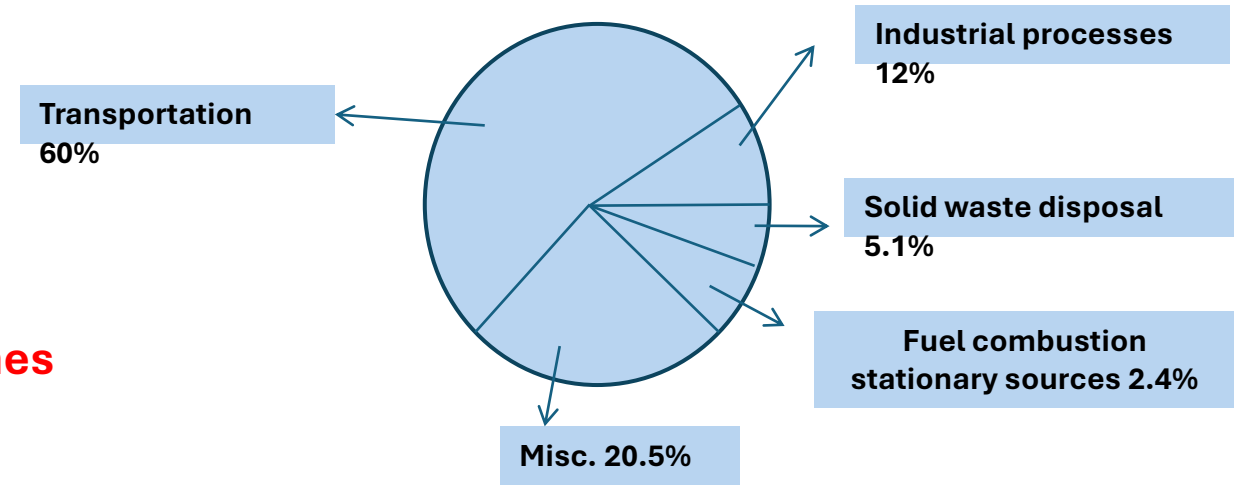


Volatile Organic Compounds (VOCs)

- A great variety of VOCs, including many HCs, enter the atmosphere from both natural and anthropogenic sources
- The petroleum industry is the main anthropogenic source of HCs in the atmosphere
- Gasoline is a complex mixture of many volatile HCs, and in urban areas, gasoline vapors can escape into the atmosphere in several ways

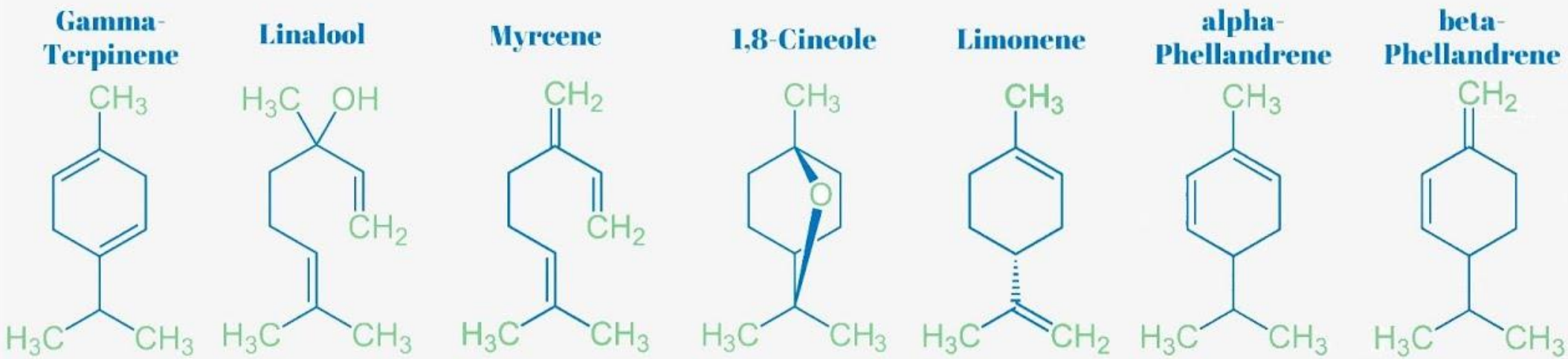
1. When gas is pumped at gas stations
2. During the filling of storage tanks
3. Unburned gasoline in exhaust

from automobiles and small combustion engines



- In the natural world, the pleasant aroma of pine, eucalyptus, and sandalwood trees are caused by the evaporation of VOCs called terpenes from their leaves

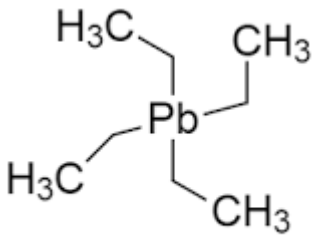
Terpenes



General formula : $(C_5H_8)_n$

What is gasoline?

The typical composition of gasoline hydrocarbons (% volume) is as follows:
4-8% **alkanes**; 2-5% **alkenes**; 25-40% isoalkanes; 3-7% **cycloalkanes**; l-4% cycloalkenes; and 20-50% total **aromatics** (0.5-2.5% benzene)



Structure of Tetraethyl Lead

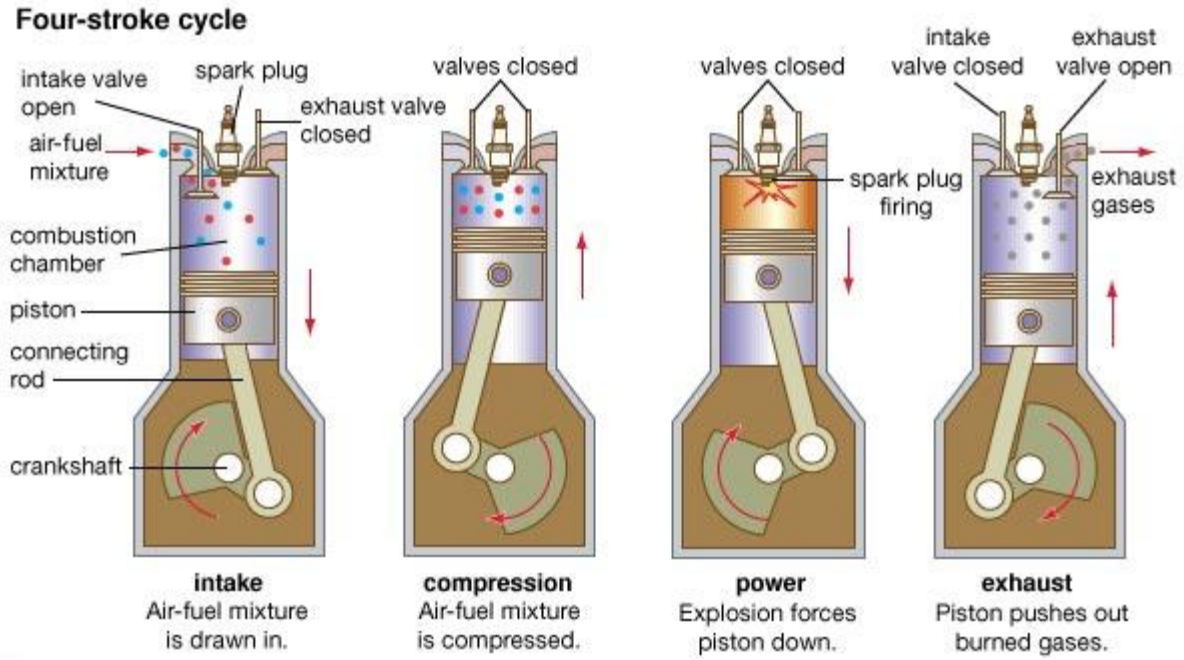
Tetraethyl lead acts as an anti knocking agent in gasoline and jet fuel and is added to petrol in order to reduce the ignition of vapors of petrol. Thus, tetraethyl lead is a petroleum additive.

Knocking occurs when fuel burns unevenly in your engine.

Volatile Organic Compounds (VOCs)

❖ Automobile Four-Cycle Internal Combustion Engine

- Gasoline-powered four-cycle internal combustion engines are predominant in more developed countries
- To know how NO_x, CO, and HCs are produced through this, we need to learn how this engine works
- There are four steps in one complete cycle of the gasoline engine
- In the confined space of the internal combustion engine, oxygen is in limited supply, and CO is formed and released to the atmosphere



Volatile Organic Compounds (VOCs)

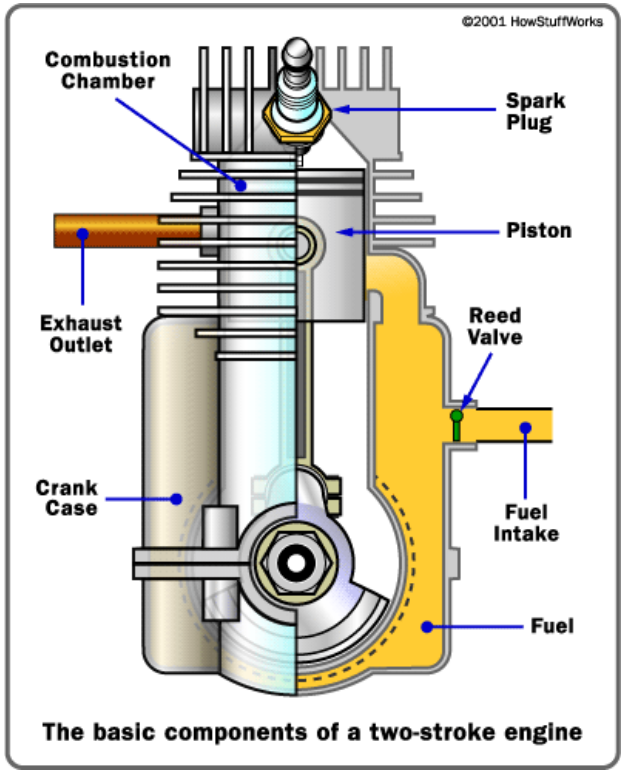
❖ Automobile Four Cycle Internal Combustion Engine

- In the internal combustion engine, temp and pressure are very high, and NO is formed from N_2 and O_2. The unburned HCs are emitted with the burned gas as pollutant VOCs
- The air/fuel ratio has a dramatic effect on the emission of pollutants from the four-cycle engine
- During the engine tune-up, adjustment of the carburetor or the fuel injection system can greatly reduce the emission of pollutants.
- The ideal stoichiometric air/fuel ratio is 14.7:1
- Adjustment to a more fuel-rich (lower air/fuel) ratio makes the car easier to start and decreases the emission of NOx and HCs; unfortunately, it also decreases the fuel efficiency, and the car will get fewer Km per liter. The emission of CO is also increased as air is less available
- Adjustment to a more air-rich (higher air/fuel) ratio will make the car harder to start; although it increases fuel efficiency, the emission of NO_x is increased
- The ideal ratio is a compromise between fuel efficiency and the emission of the lowest collective amount of pollutant

Volatile Organic Compounds (VOCs)

- **Gasoline Powered Two-Cycle Engine**
- **A two-cycle engine is used to power motor scooters, mopeds, and snowmobiles. It is less complicated than the four-cycle engine and also less expensive.**
- **It takes fuel, releases exhaust in the same stroke, and emits from 25% to 30% of the fuel consumed as unburned HCs.**
- **The two-cycle engine has no valves, and that power is produced in one of two steps rather than one in four steps; this means the two-cycle engine runs faster and hotter than the four-cycle engine**

1. In two-cycle engine motor oil: gasoline 1: 40 ratio
2. Motor oil has C_{18} to C_{25} HCs i.e., higher molecular weight
3. In the two-cycle engine, combustion takes place at a lower temperature than the four-cycle engine
4. Due to higher molecular weight, motor oil is not vaporized and burned efficiently like gasoline
5. Two step cycle requires that the exhaust gases from combustion leave the cylinder through the exhaust port while the fresh air/fuel mixture for the next power stroke is simultaneously enters the cylinder from the side arm



The basic components of a two-stroke engine

Volatile Organic Compounds (VOCs)

❖ **Comparison**

1. A comparison of the emission from marine outboard two- and four-cycle engines of the same horsepower was made. The result of those tests show that the exhaust of a two-cycle engine contained more than 12 times the amount of HCs than a four-cycle engine of the same power

Type of Marine outboard Engine	CO(g)	NO_x(g)	HC(g)
Two cycle Engine	165	0.3	89
Four Cycle Engine	127	0.7	7
Source: Juttner F. D. et al., Water Research **1995**, *29*, 1976-1982			



Automobile Pollutants

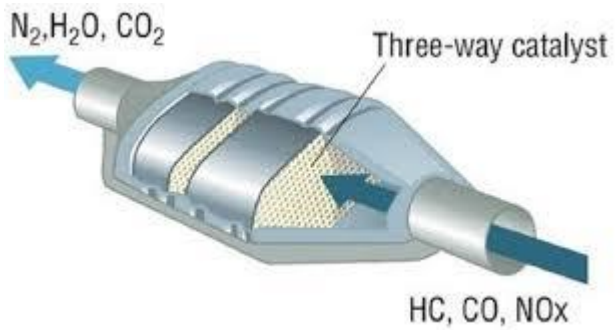
- **Motor vehicles are a major source of CO, NO_x, and volatile HCs**
- **Since 1975, when all new cars in the USA were required by law to be equipped with a catalytic converter, emissions of those pollutants have been reduced significantly**
- **The table shows that today's car emits 95% less pollutants than pre-1970 vehicles despite the fact that the number of miles traveled has almost doubled in the last 20 years**

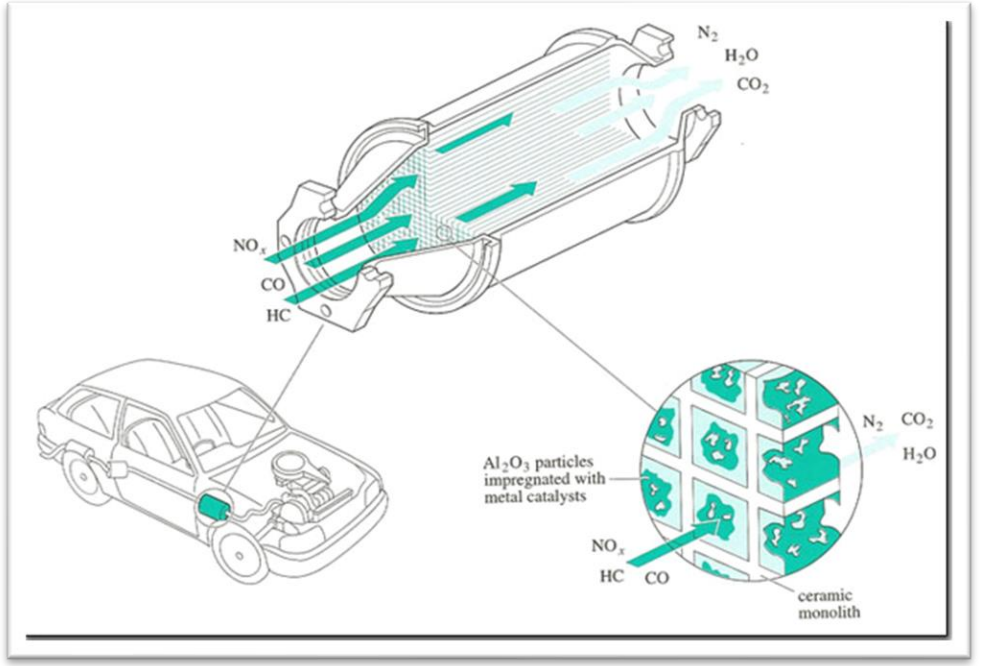
Emission Standards for Light duty Vehicles								
	Federal Standards				California Standards			
Year	HCs	CO	NOx	Evaporative HCs	HCs	CO	NOx	Evaporative HCs
<1970	10.6	84	4.1	> 45	10.6	84	4.1	>45
1970	4.1	34	-	-	4.1	34	-	2
1975	1.5	15	3.1	2	0.9	9	2.0	2
1980	0.41	7.0	2.0	2	0.39	9	1.0	2
1985	0.41	3.4	1.0	2	0.39	7	0.4	2
1990	0.41	3.4	1.0	2	0.39	7	0.4	2
1993	0.41	3.4	1.0	2	0.25	3.4	0.4	2
2000	0.41	3.4	0.4	2	0.25	3.4	0.4	2
All values reported in gms per mile except for evaporative HCs which are expressed as gms per tests								

The Catalytic Converter

- The use of a three-way catalytic converter (used since 1981) can reduce the emission drastically

- It is called a three-way catalytic converter as it simultaneously reduces the amounts of HCs, NO_x, and CO in the exhaust stream
- The converter is a very fine honeycomb structure made of ceramic coated with precious metals like Pt, Pd, and Rh, which act as catalysts





Top Five Automotive Catalytic Converter Market Vendors

BASF SE

CDTi Advanced Materials Inc.

Continental AG

Eberspacher Group

Faurecia SA

The Catalytic Converter

- The catalytic converter has two chambers in succession. As the gases enter, **Rh** catalyzes the reduction of NO_x to nitrogen gas by hydrogen generated at the surface of the **Rh** catalyst by the reaction of H_2O and unburned HCs

$$HCs + H_2O \longrightarrow H_2 + CO \; ; \qquad 2NO + 2H_2 \longrightarrow N_2 + 2H_2O$$

- Then air is injected into the exhaust stream to introduce oxygen and **CO** is oxidized to **CO_2** in the presence of **Pt** and **Pd** catalyst

$$2\,CO + O_2 \longrightarrow 2\,CO_2 \; ; \quad HC + 2O_2 \longrightarrow CO_2 + 2\,H_2O$$

- Thus, overall reaction for reduction of NO and oxidation of CO can be written as

$$2\,NO + 2\,CO \xrightarrow{\textbf{Rh, Pt, Pd Cat}} N_2 + 2\,CO_2$$

The Catalytic Converter

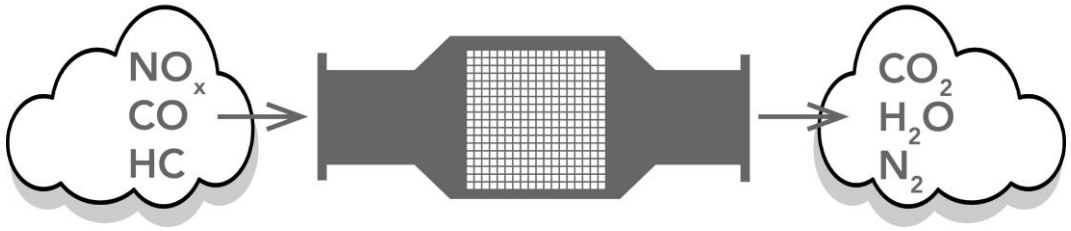
- Oxidation of a typical gasoline HC, Octane occurs as

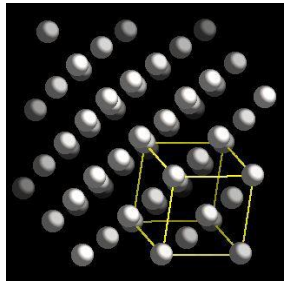
$$2\,C_8H_{18} + 25\,O_2 \xrightarrow{\text{Pt, Pd Cat}} 16\,CO_2 + 18\,H_2O$$

- Automobiles using catalytic converters must have their air/fuel ratio set as 14.7:1 to get optimum results.

- At room temp, the efficiency of the catalytic converter is nearly zero



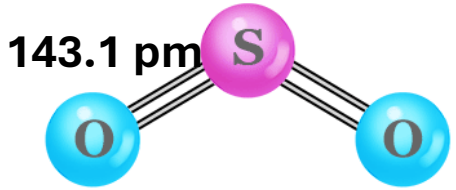


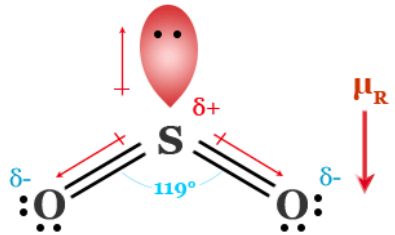


Pt, Rh, and Pd (FCC)

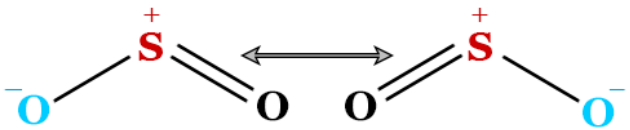
Surface chemistry

Sulfur Dioxide





dipole moment of sulphur dioxide
(μ = 1.61 D)



Resonance structures of sulfur dioxide

- The release of SO_2 into the atmosphere is the primary cause of acid rain
- Fossil fuel combustion accounts for 70% of the emission
- Industrial sources contribute approximately 23% of SO_2
- Coal oil and all other fossil fuels naturally contain some sulfur (FeS).

- When sulfur containing coal is burned, the sulfur is oxidized to SO_2: $S + O_2 \longrightarrow 2SO_2 + 2H_2O$
- H_2S (produced as an end product of anaerobic decomposition of sulfur containing organic matter by microorganism) enter to the atmosphere to form SO_2

$$2H_2S + 3O_2 \longrightarrow 2SO_2 + 2H_2O$$

Sulfur Dioxide





- Volcanic eruptions are another more localized natural source of SO_2; e.g., the eruption of Mt. Pinatubo in the Philippines in June 1991 contributed 25 million tons of SO_2 into the atmosphere, where it was converted into sulfuric acid aerosols

Fate of atmospheric SO_2: Acid Rain

- The residence time of SO_2, controlled not only by chemical conversion to sulfate but also by dry deposition, is 10-30 days.

- SO_2 in the atmosphere reacts with oxygen to form SO_3, which then readily reacts with water vapor or water

The fate of atmospheric SO_2: Acid Rain

$$SO_2 + OH\cdot \longrightarrow HSO_3\cdot$$

$$HSO_3\cdot + O_2 \longrightarrow SO_3 + HOO\cdot$$

$$SO_3 + H_2O \longrightarrow H_2SO_4\ (g)$$

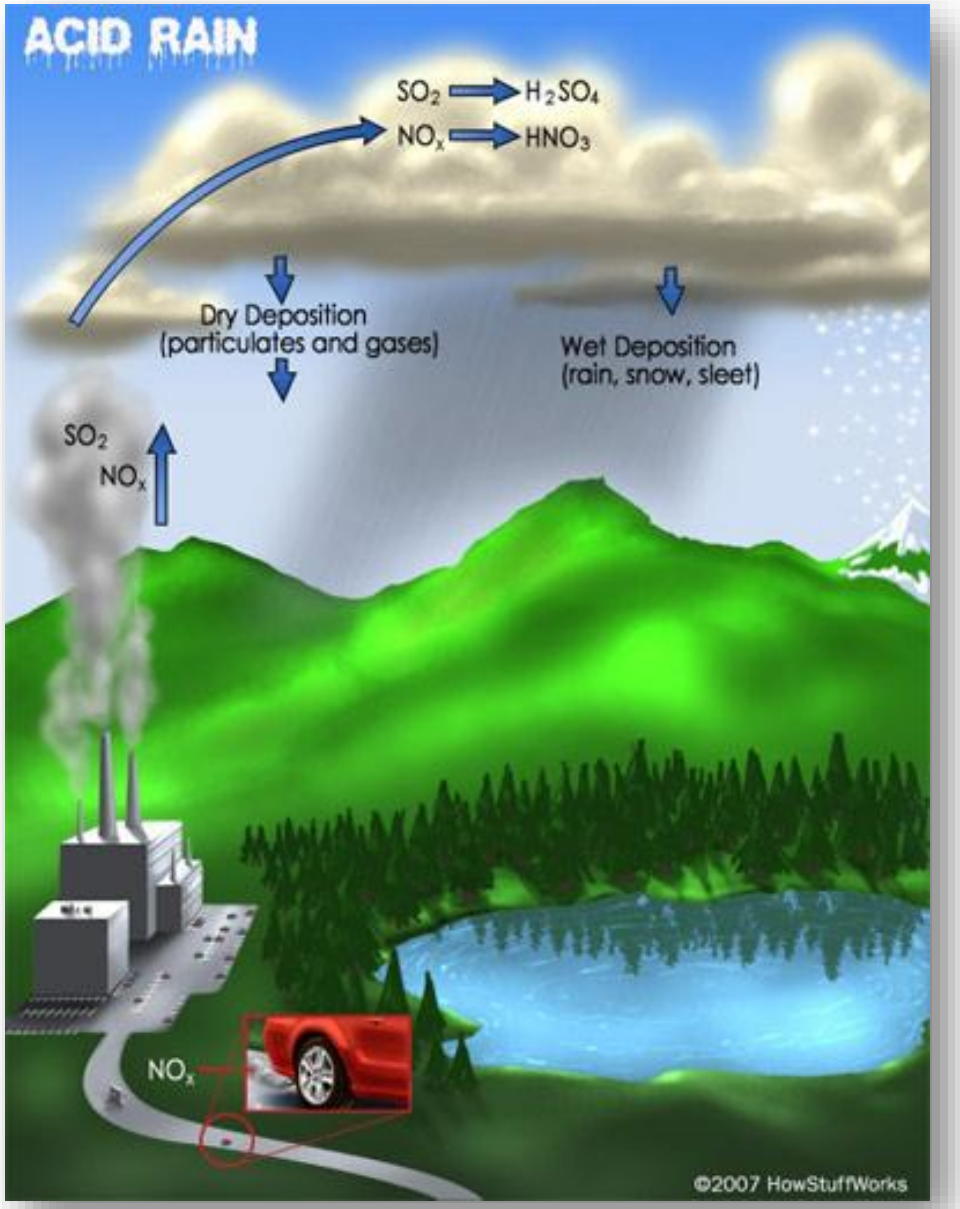
$$H_2SO_4 + H_2O \longrightarrow H_2SO_4\ (aq)$$

- Sulfuric acid in the atmosphere becomes concentrated near the base of clouds where the pH level is as low as 3 (pH of orange juice)
- Some of the atmospheric SO_2 dissolves if there is significant water in the air to form sulfurous acid

$$SO_2(g) + H_2O\ (aq) \longrightarrow H_2SO_3(aq)$$

- The dissolved SO_2 is oxidized by trace amount of H_2O_2 and Ozone that are also present in the aerosol droplets to sulfate ions (SO_4^{2-})
- $HSO_3^- + H_2O_2 \longrightarrow H_2O + HSO_4^-$

$HSO_3^- + O_3 \longrightarrow O_2 + HSO_4^-$



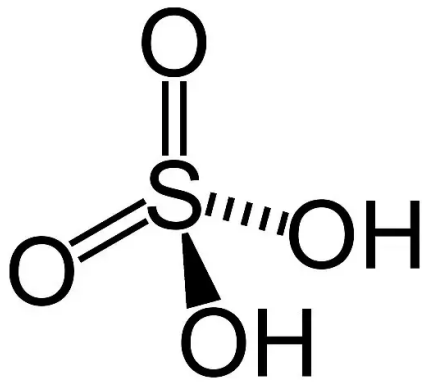
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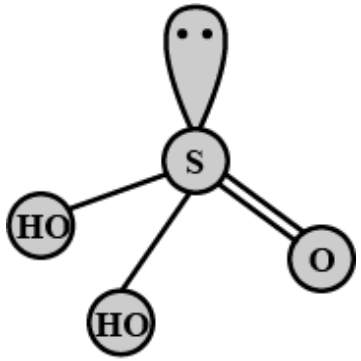
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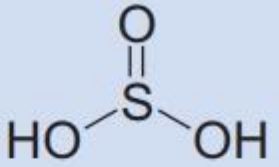
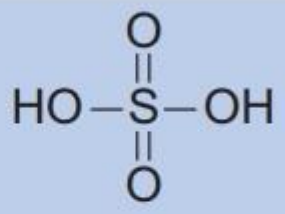
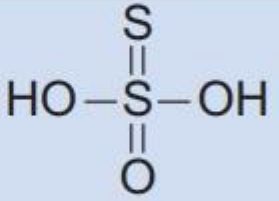
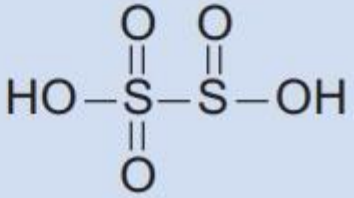
$HSO_3^- + O_3 \longrightarrow O_2 + HSO_4^-$

H_2SO_4 *vs* H_2SO_3

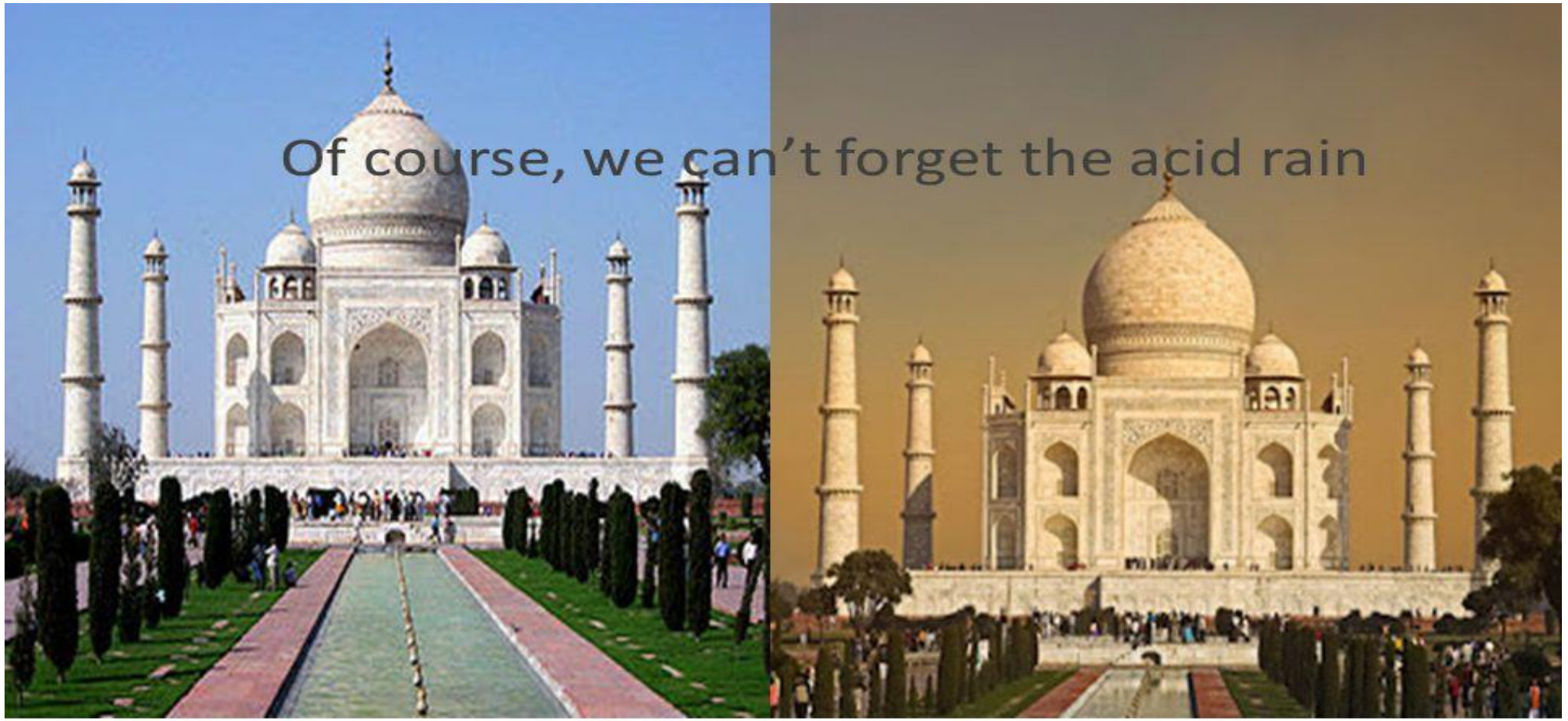


Sulphurous acid
(H_2SO_3)

Oxoacids of Sulfur

Name	Molecular Formula	Structure
Sulphurous acid	H_2SO_3	
Sulphuric acid	H_2SO_4	
Thiosulphuric acid	$H_2S_2O_3$	
Dithionous acid	$H_2S_2O_4$	
Dithionic acid	$H_2S_2O_5$	

Effect of Acid Rain



Original color of Taj Mahal

Taj Mahal after the constant effect of acid rain

Effect of SO_2 on Human Health and Environment

- SO_2 is a colorless, toxic gas with a sharp, acidic odour
- Exposure to it causes irritation of eyes, respiratory passages, and aggravates symptoms of respiratory disease. Children are susceptible to its effects
- SO_2 is also harmful to plants. Crops such as barley, alfalfa, cotton, and wheat are particularly adversely affected



Barley



Alfalfa



Cotton



Wheat

Effect of SO_2 on Human Health and Environment

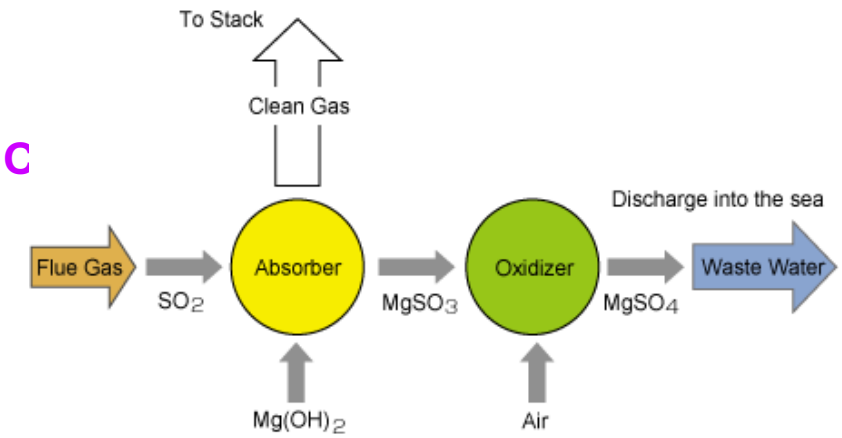
❖ How to control SO_2 emissions

1. **Sulfur can be removed from coal before combustion**
2. **SO_2 can be removed from the smokestack after combustion but before it reaches to atmosphere**

- ❖ **The second approach is cheaper and most common**

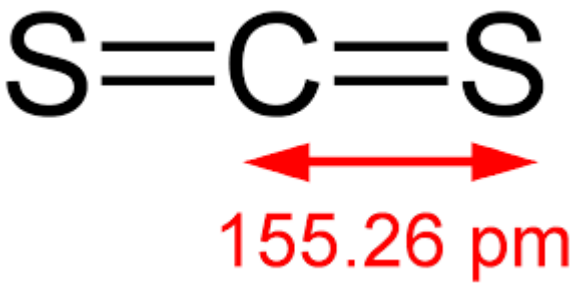
- ❖ **The most commonly used method is flue gas desulfurization (FGD), in which sulfur-containing compounds are washed out by passing the Chimney (flue) gases through a slurry of water mixed with finely ground limestone ($CaCO_3$) or dolomite [$Ca•Mg(CO_3)_2$] or both.**

- ❖ **On heating the basic $CaCO_3$ with acidic SO_2 in the presence of O_2 to form C**



$$2\,SO_2 + 2\,CaCO_3 + O_2 \longrightarrow 2\,CaSO_4 + 2CO_2$$

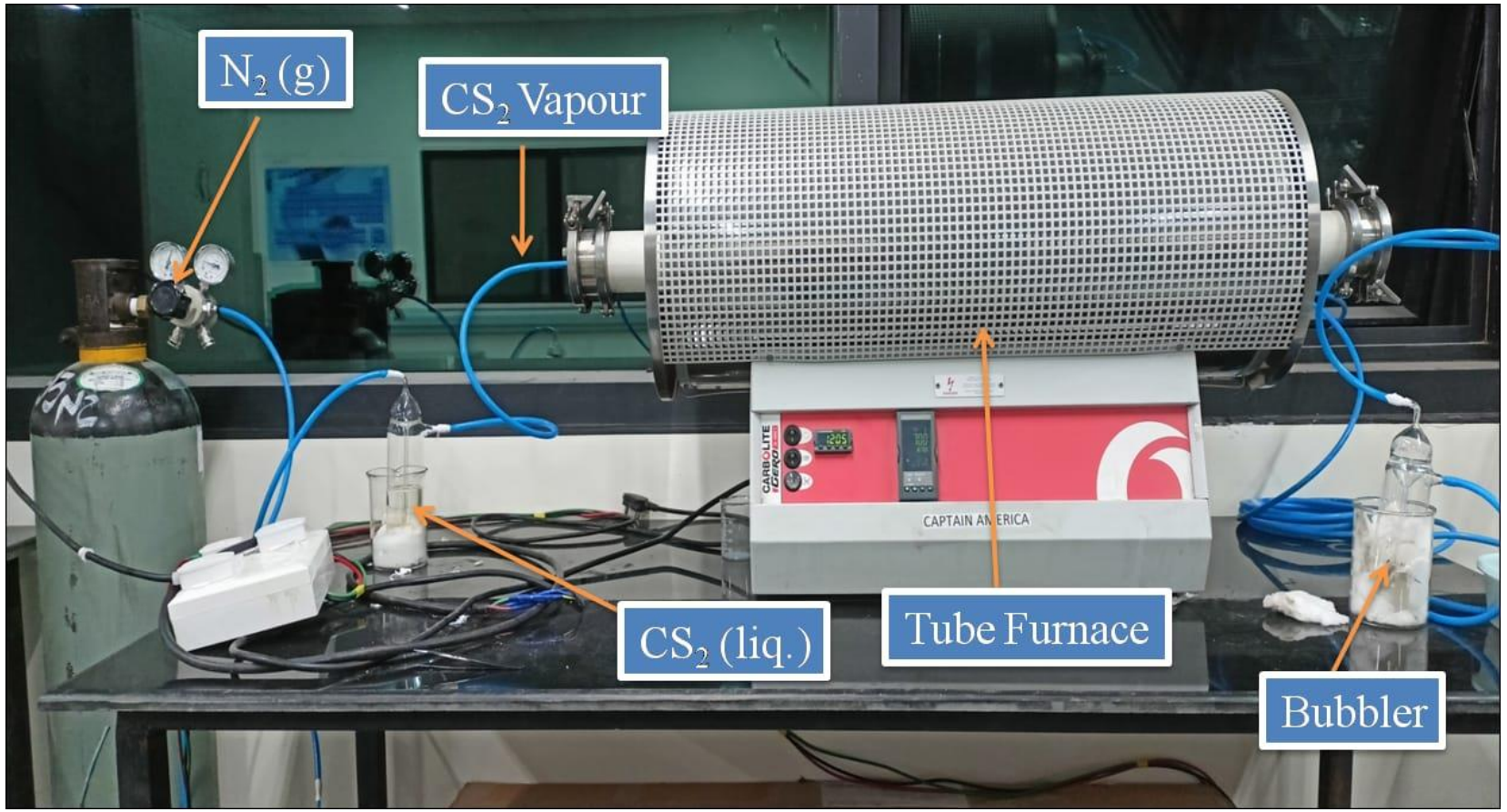
Flue Gas: a mixture of gases produced by the burning of fuel or other materials in power stations and industrial plants and extracted via ducts.



a colorless liquid with an ether-like odor

Boiling point: 46.3 °C

$CS_2(s) + O_2(g) = CO_2(g) + SO_2(g)$

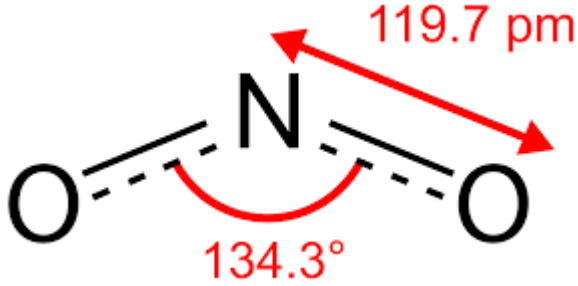


Industrial Smog

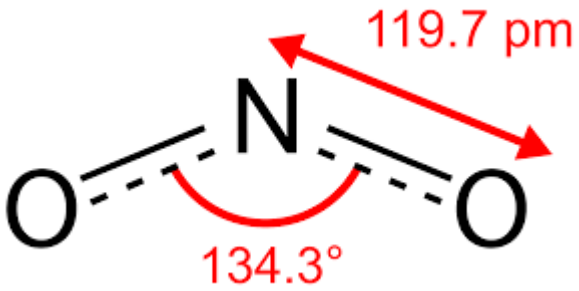
- Particulate matter and **SO_2** can be a deadly combination. Released into the atmosphere together when coal is burned, they can form **industrial smog:** a mixture of fly ash, soot, SO_2, and some VOCs
- It is formed in winter, typically in cities where the weather is cold and wet. Visibility was often reduced to a few yards and people in factory towns lived under a pall of black smoke

❖ Photochemical Smog

- The ***origin of photochemical smog is quite different from that of industrial smog***. Typically, photochemical smog develops as a yellow-brown haze in hot summer weather in cities where automobile traffic is congested.
- The reaction that led to its formation is initiated by sunlight and involves the HCs and NO_x emitted in automobile exhaust.
- NO_2 is responsible for the brownish color of the haze



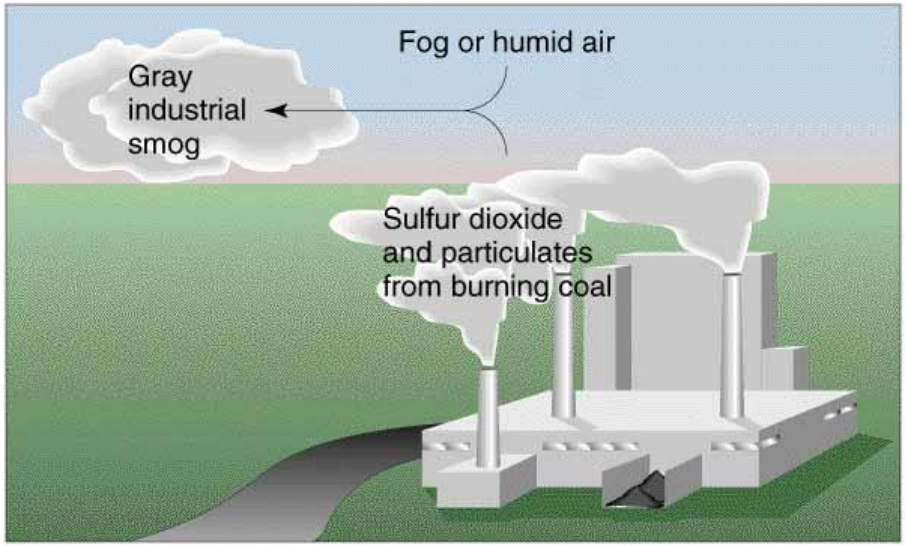
Industrial Smog



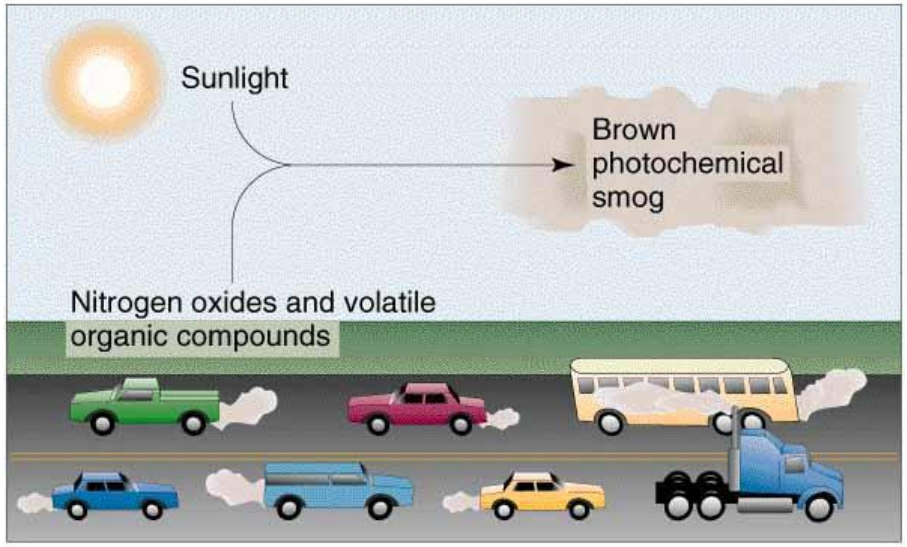


NO_2 transforms to the colorless dinitrogen tetroxide (N_2O_4) at low temperatures (below −11.2 °C) and reverts to NO_2 at higher temperatures.

Industrial Smog



(a) Industrial smog



(b) Photochemical smog





Production of Hydroxyl Radicals

- By the time sunlight reaches the surface of the Earth, all of the light energy UV light has been absorbed in the stratosphere
- NO_2 is the only automobile emission that is capable of absorbing visible light that reaches the Earth's surface
- $NO_2 + \text{sunlight } (< 320 \text{ nm}) \longrightarrow NO + O$; $O + O_2 \longrightarrow O_3$
- $O_3 + NO \longrightarrow NO_2 + O_2$

- The ozone produced absorbs light in the blue region of the visible spectrum (< 320 nm) and photo dissociates:

  $O_3 \longrightarrow O_2 + O$
- The oxygen atom produced (having six electrons) reacts with water vapor in the atmosphere and abstracts a hydrogen atom to produce hydroxyl radical (seven electrons)
- $O + H_2O \longrightarrow 2OH^{\bullet}$ By this way one NO_2 molecule produces two hydroxyl radicals
- The concentration of hydroxyl radicals does not continue to increase out of control because there are termination reactions that remove it from the troposphere. It can react with other radical species in the troposphere

Production of Hydroxyl Radicals

$$\cdot OH + \cdot NO_2 \longrightarrow HNO_3$$

$$\cdot OH + HOO\cdot \longrightarrow H_2O + O_2$$

$$2\,\cdot OOH \longrightarrow H_2O_2 + O_2$$

❖ The products are very soluble in water and are removed from troposphere during precipitation

➢ Unburned HCs in automobile exhausts (RCH_3) react with hydroxyl radicals to form a number of secondary pollutants, including HC radicals $RO_2\bullet$. This radical then reacts with NO to form aldehydes and the hydroperoxide radical $HO_2\bullet$

$$RCH_3 + \cdot OH \longrightarrow RCH_2\cdot + H_2O$$

$$RCH_2\cdot + O_2 \longrightarrow RCH_2OO\cdot \quad \text{(Peroxy alkyl radical)}$$

$$RCH_2OO\cdot + NO \longrightarrow RCH_2O\cdot \quad \text{(Alkoxy radical)}$$

$$RCH_2O\cdot + O_2 \longrightarrow RCHO + \cdot OOH \quad \text{(Hydroperoxyl radical)}$$

$$HOO\cdot + NO \longrightarrow NO_2 + \cdot OH$$

Production of Hydroxyl Radicals

❖ Each step in this reaction produces a radical. The overall reaction can be summarized as

$$RCH_3 + 2\,O_2 + 2\,NO \longrightarrow RCHO + 2\,NO_2 + H_2O \qquad (1)$$

$$2\,NO_2 + 2\,H_2O \longrightarrow 2\,NO + 4\,\dot{O}H \qquad (2)$$

$$RCH_3 + 2\,O_2 + H_2O \longrightarrow RCHO + 4\dot{O}H$$

❖ This reaction produces four hydroxyl radicals for every HC reacted. This is a catalytic reaction.

A very small number of radicals can produce a large amount of product through the production of four radicals per cycle

Reaction of Hydroxyl Radicals with HCs

❖ Abstraction of Hydrogen

➢ Hydroxyl radicals will react with certain unburned HCs from the automobile exhaust depending on the number and type of C-H bonds in the HCs

Reaction of Hydroxyl Radicals with HCs

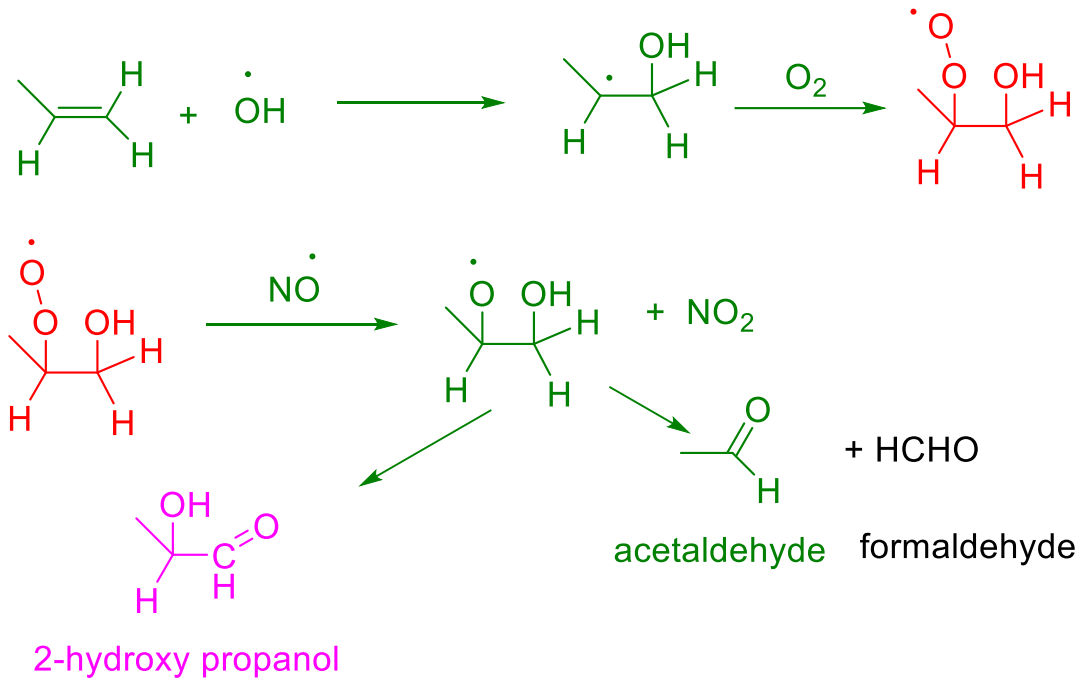
Compound	Bond	Energy KJ/mol
Methane	CH_3-H	427
Ethane	CH_3-CH_2-H	406
Propane	CH_3-CH_2-CH_2-H	393
Methanol	$HOCH_2$-H	393
Benzene	H_5C_5C-H	427
Toluene	$H_5C_6H_2$C-H	326

- The dissociation reaction is $R\text{-}H \longrightarrow R^{\bullet} + H^{\bullet}$
- The bond dissociation energy depends on how the R radical is stabilized $3^o > 2^o > 1^o$ i.e., a tertiary C-H bond more easily dissociate than a secondary C-H bond
- For benzene, the C-H has high dissociation energy as the aromatic carbon is sp^2, which mean sp^2 C-H bonds are much stronger than aliphatic C-H bond due to higher *s*-character
- Higher reaction rates (low dissociation energy) are observed for toluene or xylene as the $C_6H_5CH_2\bullet$ can be stabilized by adjacent aromatic ring

Reaction of Hydroxyl Radicals with HCs

❖ Addition to the double bond

- The reaction of hydroxyl radicals with alkenes proceeds at even a faster rate than the hydrogen abstraction for HCs
- This reaction is not a hydrogen atom extraction but rather an addition of the hydroxyl radical to the double bond



- The NO_2 produced in this reaction can go on to make more ozone, and the products of this addition reaction i.e., acetaldehyde, formaldehyde, and 2-hydroxypropanol all go on to form other pollutants in secondary smog-forming reactions

Secondary Smog Forming Reactions

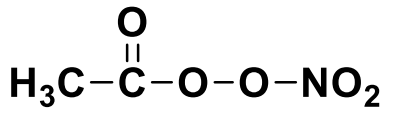
- The products formed in the previous reactions can undergo further reactions to the troposphere. The following reaction sequence takes place for all aldehydes formed. Acetaldehyde is used as example

$$CH_3CHO + \cdot OH \longrightarrow CH_3CO\cdot + H_2O$$

$$CH_3CO\cdot + O_2 \longrightarrow CH_3COOO\cdot$$

$$CH_3COOO\cdot + \cdot NO_2 \longrightarrow CH_3COOONO_2$$

PAN



Peroxyacetyl nitrate

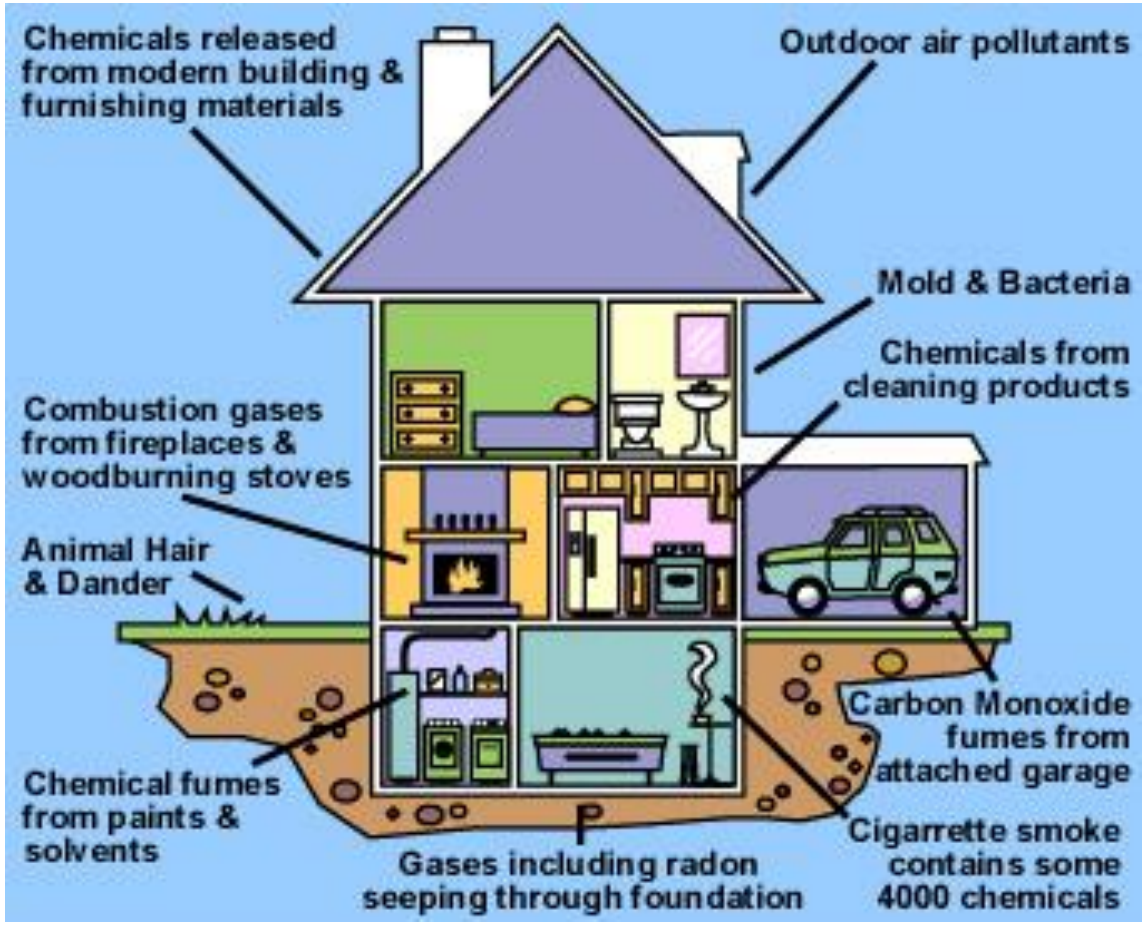
- PAN is the component of smog that causes major eye irritation. PANs are relatively stable molecules and have long lifetimes in cooler air
- Ozone, aldehyde, and PANs all contribute to the harmful effects of photochemical smog, but ozone the pollutant produced in the greatest quantity causes the most serious problems in the troposphere
- Ozone in the stratosphere protects us from damaging UV radiation from the Sun
- Ozone is a powerful oxidizing agent and causes irritation in the eyes and nasal passages
- Ozone is also very toxic to plants

Indoor Pollution

➢One can expect to be safer from air pollutants indoors, but in today's well-sealed homes and offices, this is often not the case

➢In buildings where there is little or no circulation or fresh air, pollutants may accumulate to a dangerous level

Lead pipes

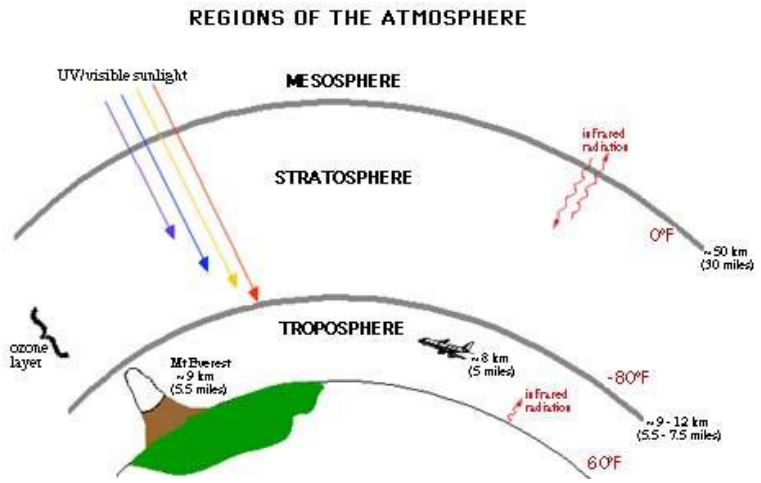


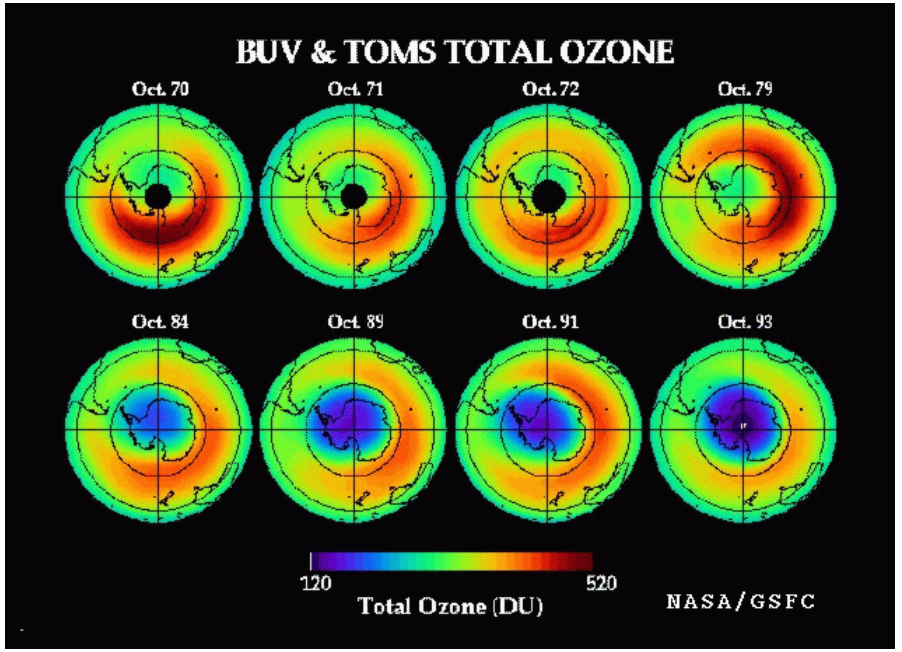
Chemistry of the Stratosphere

- Today, the ozone layer is considered the Earth's natural sun-screen because it filters harmful UV radiation before it can reach the surface of the Earth
- A substantial reduction in the amount of ozone in the ozone layer could threaten all life on Earth

Dobson Unit

- The Dobson Unit (DU) is used to describe the amount of ozone in the stratosphere
- It is named after Gordon M. B. Dobson, who built the first instrument to measure the total abundance of ozone from the ground
- **1 DU = 2.7 x 10^{16} ozone molecules/sq centimeter**
- In the past, the average amount of ozone covering the Earth was more than 270 DU
- In 2000, the average abundance of ozone from 90° N to 90° S was 293.4 DU
- However, the amount of ozone over the Antarctic is 220 DU which is refer to the ozone hole

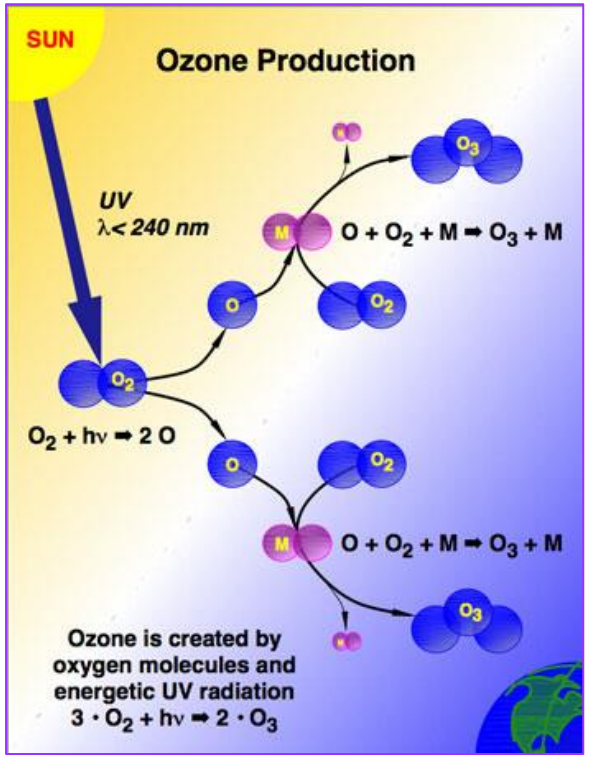




The production of Ozone in the Stratosphere

- In the absence of pollution, ozone (O_3), is not present in any appreciable amount in the troposphere but occurs naturally in the stratosphere
- Ozone's concentration is greatest at an altitude of 20 to 30 km from the Earth's surface
- The ozone layer is formed when an ordinary molecule of oxygen gas (O_2) in the stratosphere absorbs UV radiation from the sun wavelength less than 240 nm and this dissociates O_2 to a single oxygen atom (O)

O_2 + hv (< 240 nm) ⟶ O + O (slow) $\Delta H = 498$ KJ/mol



- Single oxygen atoms are very reactive and immediately combine with **O_2** to form **O_3:**
- **$O + O_2 + M \longrightarrow O_3 + M + Heat$** (fast)
- M can be a third molecule (N_2) present in the atmosphere
- The heat generated by this reaction is carried away by the third molecule
- At the same time, the ozone formed absorbs UV radiation very strongly with wavelengths of 220-330 nm to be broken down into an oxygen molecule and oxygen atom
- Also, when O_3 encounters an oxygen atom, two can combine to form two O_2 molecules
- A dynamic equilibrium is established

The production of Ozone in the Stratosphere

Common names for the UV spectrum			
l(nm)	Name	Species absorbing	Location
10-240	Far UV	O_2, N_2	Thermosphere
250-290	UV-C	O_3	Stratosphere
290-320	UV-B	O_3	Stratosphere, troposphere
320-380	UV-A	NO_2	Polluted troposphere
400-750	Visible	Many	Earth's Surface

Catalytic destruction of Ozone

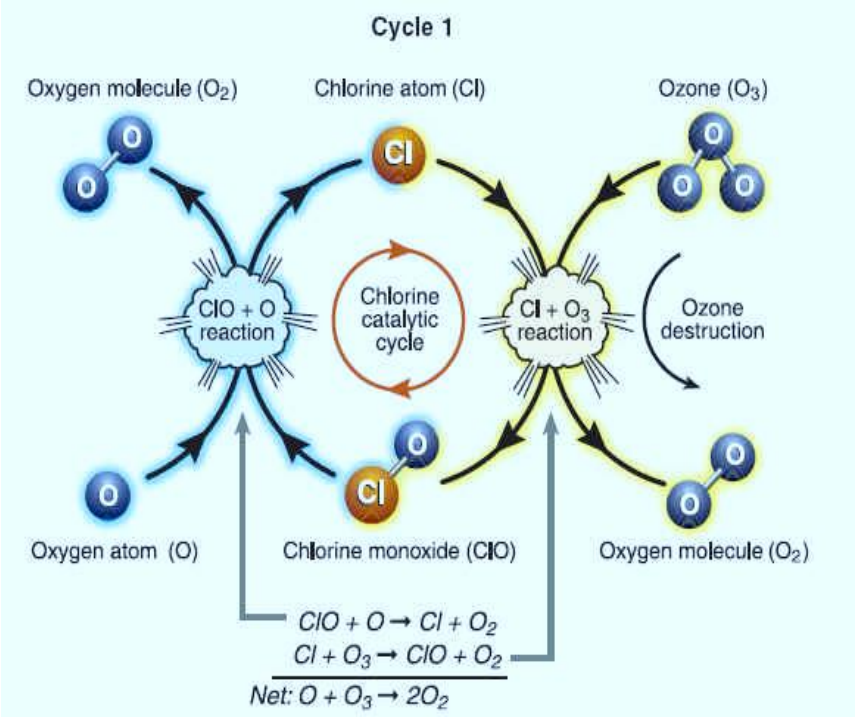
- Ozone is a very reactive molecule and is termed a meta-stable molecule i.e. it decomposes slowly in contact with a molecule of another gas
- There are a number of species that react efficiently by abstracting an oxygen atom from the ozone molecule. Say "X" designates the reactive species. Thus, the reaction steps are

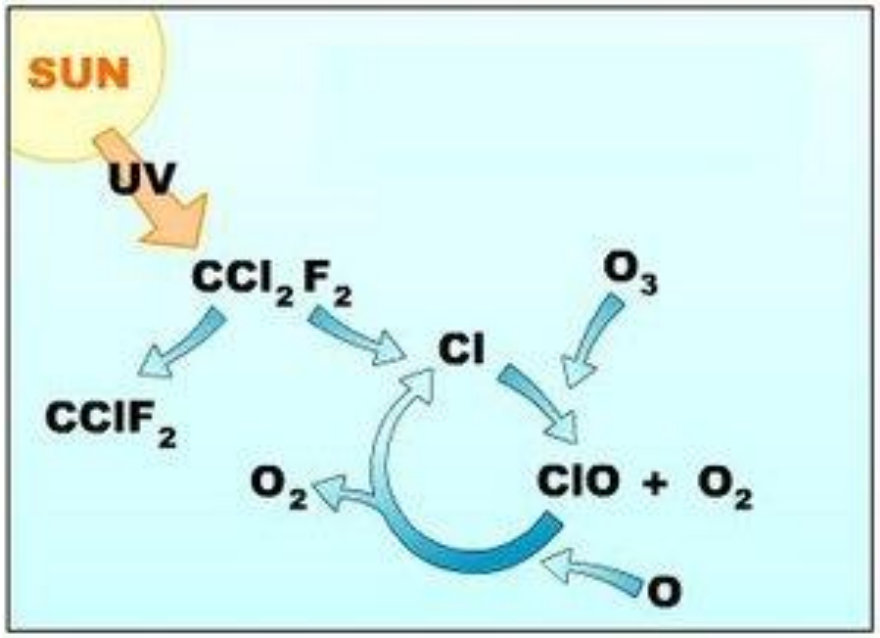
  i) $X + O_3 \longrightarrow XO + O_2$; ii) $XO + O \longrightarrow X + O_2$

- These are the additional steps for ozone destruction compared to the steps discussed earlier

The production of Ozone the Stratosphere

- So, the sum of above two reactions is $O_3 + O \longrightarrow 2O_2$
- The overall reaction does not contain the X species, because X is not consumed in the reaction. It acts as a catalyst for the destruction of ozone
- "X" lowers the activation energy and atomic oxygen is required to regenerate X
- The catalytic species "X" can be i) Hydroxyl radical, ii) nitric oxide, iii) Chlorine , and iv) bromine atoms





DESTRUCTION OF OZONE BY CFC

Catalytic destruction of Ozone

Hydroxyl Radical Cycle

- Hydroxyl radical ($^{\bullet}OH$) can be produced by two different photochemical processes. The first is hydrogen abstraction with either water or methane

  i) $O + H_2O \longrightarrow 2\,{}^{\bullet}OH$; ii) $O + CH_4 \longrightarrow {}^{\bullet}OH + {}^{\bullet}CH_3$

- The second is photolysis of **water** $H_2O + hv \longrightarrow H^{\bullet} + {}^{\bullet}OH$
- Hydroxyl radical is responsible for nearly one-half of the total ozone destruction in the lower stratosphere

$$^{\bullet}OH + O_3 \longrightarrow {}^{\bullet}OOH + O_2$$

$$^{\bullet}OOH + O \longrightarrow {}^{\bullet}OH + O_2$$

Net reaction $O + O_3 \longrightarrow 2O_2$

One $^{\bullet}OH$ radical can have as many as 40 cycles i.e., 40 ozone molecule can decompose

Catalytic destruction of Ozone

Nitric Oxide Cycle

- Auto mobiles and truck engines release large quantities of nitric oxide (NO) into the troposphere
- Nitric oxide is converted to NO_2 and eventually to HNO_3 which fall down as rain fall before reaching to stratosphere
- N_2O is much less reactive than NO and reaches to stratosphere
- Above 30 km, the N_2O can absorb high energy photons to produce molecular nitrogen and an excited oxygen atom (O*)

  $N_2O + hv \rightarrow N_2 + O^*$

- Below 30 km in the stratosphere, the excited state oxygen reacts with the N_2O to produce NO

  ii) **$N_2O + O^* \rightarrow 2\,NO^\bullet$**

- NO can act as X in the catalytic process

1 NOx radical can catalyze the destruction of 10,000 ozone molecules

$$NO + O_3 \longrightarrow NO_2 + O_2$$

$$NO_2 + O \longrightarrow NO + O_2$$

$$\text{Net } O + O_3 \longrightarrow 2O_2$$

Catalytic destruction of Ozone

The Chlorine Cycle

- The major sources of stratospheric chlorine and bromine have been the anthropogenic **chlorofluoro carbons (CFCs)** and bromine-containing halons (haloalkanes)
- CFCs used as refrigerants, propellants for aerosol spray, and solvents for cleaning electric circuit
- Halons are used in commercial fire extinguisher system
- CFCs and halons are nontoxic and nonflammable, which make them superior to other gases that were used for the same application
- CFCs are so unreactive they don't break down when released and can persist in the troposphere for more **than 100 years.** Over time, air currents carry them into the stratosphere
- In 1974, two chemists, F. S. Rowland and Mario Molina, predicted that when exposed to UV radiation in stratosphere, CFCs would break down to form chlorine radical ($Cl^{\bullet}$)
- Each chlorine radical involved in a chain reaction has the potential to destroy 100,000 molecules of ozone before winds carry it back to the troposphere
- Bromine-containing compounds are more efficient in destroying the ozone